

ACC SCIENTIFIC STATEMENT

Antiplatelet Therapy in the Management of Atherosclerotic Cardiovascular Disease: 2026 ACC Scientific Statement



A Report of the American College of Cardiology

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ABSTRACT

Antiplatelet medications have been fundamental to the management of atherosclerotic cardiovascular disease (ASCVD); however, recommendations for their use have evolved over the past decade. Aspirin remains the most widely used antiplatelet drug. Recent data do not support a benefit for primary prevention of ASCVD in unselected populations with routine aspirin use; consideration should be reserved for those at highest ischemic risk. Similarly, the evolution of drug-eluting stents, antiplatelet medications, and research on the optimal regimen and duration of dual antiplatelet therapy have altered evidence-based recommendations for use of antiplatelets in managing patients diagnosed with ASCVD. Antiplatelet medications remain powerful tools to prevent ischemic events, particularly for those who have experienced an acute coronary syndrome, but the balance between the risk of ischemia and the risk of bleeding poses a perpetual challenge and requires individualized treatment.

This scientific statement provides a detailed overview of current evidence for the use of antiplatelets in the management of ASCVD. This includes a comparison of current clinical practice guideline-based recommendations, summary of pivotal trials, and discussion of antiplatelet use in patients with elevated bleeding risk or those undergoing surgery. Considerations for use of antiplatelets in those with different phenotypes of ASCVD and those with indications for concurrent use of antithrombotic medications, as well as long-term monitoring and adherence, are addressed.

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INTRODUCTION

The American College of Cardiology (ACC) has a long history of developing documents to complement clinical practice guidelines. Among these documents, scientific statements represent a novel approach to inform clinicians about areas where the scientific evidence is new and evolving or where sufficient data are more limited. Despite a long history as a cornerstone of pharmacologic management of atherosclerotic cardiovascular disease (ASCVD), the role of antiplatelet therapy continues to evolve. Achieving an appropriate balance between the risks of ischemia and bleeding poses a perpetual challenge, requiring recommendations tailored to the individual. This scientific statement aims to provide clinicians with current, evidence-based consensus recommendations for the use of antiplatelet therapies in primary and secondary prevention of ASCVD.

To accomplish this work, the ACC convened a writing committee in December 2025 via a confidential conference call attended only by writing committee members and ACC staff. A review of seminal publications and outstanding questions was facilitated. Writing assignments were configured according to each committee member's area of expertise, and e-mail correspondence was used to provide critical review of contributed content. Differences were resolved by consensus among the writing committee. A summary of consensus recommendations can be found in [Table 1](#). The committee's work was supported only by the ACC without any commercial input. Writing committee members were all unpaid volunteers. In accordance with ACC's policy on Relationships With Industry and Other Entities, relevant disclosures for the writing committee and comprehensive disclosures for external peer reviewers can be found in [Appendixes 1 and 2](#).

DEFINITIONS AND CLASSIFICATIONS

Bleeding Academic Research Consortium (BARC) classification: A standardized classification system, widely used in clinical trials, which is used to define bleeding severity from Type 0 (no bleeding) to Type 5 (fatal bleeding). BARC Type 1 bleeding is considered mild and nonactionable, while BARC Type 3 is clinically significant bleeding, such as bleeding requiring transfusion, intracranial bleeding, or bleeding sufficient to cause decreased hemoglobin ≥ 3 to 5 g/dL. BARC Type 4 bleeding is any bleeding related to coronary artery bypass graft (CABG) surgery, or bleeding requiring massive transfusion.³

Deescalation of antiplatelet therapy: Refers to actions of reducing the antiplatelet effect by means of changes in the type, dose, or number of drugs used to inhibit platelets.⁴

TABLE 1 Summary of Consensus Recommendations

Ischemic vs bleeding risk in determining antiplatelet strategy

- Shared decision-making for use of antiplatelet therapies is centered on understanding the risk of patient-specific future ischemic events compared with bleeding risk.
- Using risk calculators such as DAPT, PRECISE-DAPT, and ARC-HBR can be helpful, but decision-making requires global assessment of future ischemic risk balanced against bleeding risk.

Aspirin for primary prevention of ASCVD events

- Aspirin may be effective in primary prevention of cardiovascular events, but is associated with increased risk of major bleeding.¹ The current lack of benefit with aspirin for primary prevention may be due to the application of other effective preventive strategies like statins.
- Low-dose aspirin might be considered for primary prevention of ASCVD events among adults aged 40–70 y who have a higher cardiovascular risk without an increased bleeding risk.^{4,1}
- Routine use of aspirin for prevention in those over age 70 y should be avoided.¹
- Available ASCVD calculators may help identify patients at high ischemic risk. If these patients are at low bleeding risk, aspirin may be considered for primary prevention. Use of aspirin in those with HBR is associated with net harm regardless of CAC score or ASCVD risk.

Antiplatelet therapy postrevascularization

- In patients with ACS undergoing PCI, prasugrel or ticagrelor are preferred as the P2Y₁₂ inhibitor of choice, with some data favoring prasugrel. Clopidogrel may be preferentially considered in patients >75 y of age due to increased risk of major bleeding with prasugrel or ticagrelor.
- The default DAPT duration post-PCI is 6 mo for CCD and 12 mo for ACS. Aspirin should be maintained in the peri-PCI period. Shortened DAPT (1–3 mo) followed by P2Y₁₂ inhibitor monotherapy can reduce bleeding without increasing ischemic events. Shortened DAPT (1–3 mo) followed by P2Y₁₂ inhibitor monotherapy can reduce bleeding without increasing ischemic events. This approach is mainly applicable to ticagrelor (more robust data) or prasugrel, not clopidogrel or aspirin unless the patient is HBR. Caution is also advised in patients with higher ischemic risk, such as those presenting with STEMI.
- Guided or unguided deescalation from either ticagrelor or prasugrel to clopidogrel 1 mo after ACS treated with PCI may be an option in selected patients to lower bleeding risk.
- DAPT reduces the risk of vein graft failure after CABG but has little effect on clinical outcomes except for increasing the risk of bleeding.

Long-term antiplatelet therapy after ACS or MI

- Beyond 1 y after ACS, emerging data suggest that clopidogrel monotherapy may offer greater ischemic protection with similar or lower bleeding risk compared with aspirin monotherapy for secondary CAD prevention.
- Intensified strategies, including prolonged DAPT, P2Y₁₂ inhibitor monotherapy, or anticoagulant-based regimens, may be appropriate in selected high-risk patients, but ischemic benefit should be consistently weighed against risk of increased bleeding. Individualized assessment of ischemic and bleeding risk remains central to long-term antithrombotic decision-making.
- Rivaroxaban 2.5 mg twice daily with aspirin 81 mg daily can be considered for long-term use in select high-risk patients with stable CAD or PAD who do not have a history of previous major bleeding or cerebrovascular accident.

Special considerations: PAD, stroke, TIA, valve disease, and patients requiring anticoagulation

- Antiplatelet therapy has an important role in the acute and chronic management of PAD and cerebrovascular disease, similar to CAD. However, compared with CAD, shorter durations of DAPT (1–6 mo) are typically recommended postrevascularization.
- Adding rivaroxaban 2.5 mg twice daily to low-dose aspirin is a reasonable long-term strategy in patients with PAD to reduce MACE and MALE, particularly among high-risk patients who have acceptable bleeding risk.
- In patients with minor ischemic stroke (NIHSS score of ≤ 3) or high-risk TIA (ABCD² score of ≥ 4), early initiation of DAPT with aspirin plus clopidogrel within 12–24 h, continued for 21 d, reduces early recurrent stroke compared with aspirin alone, after which therapy is deescalated to antiplatelet monotherapy to limit bleeding risk. The use of DAPT among patients who receive thrombolysis or have more severe strokes (NIHSS score of >5) is unclear.

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TABLE 1 Continued

- Triple therapy among patients needing OAC and antiplatelet therapy should be avoided. Aspirin can typically be discontinued within 1 mo of PCI; clopidogrel and DOACs are the favored agents for antithrombotic therapy in these patients. Long-term management (beyond 1 y) can include DOACs as monotherapy.
- Patients receiving bioprosthetic aortic or mitral valves can be managed with single antiplatelet therapy (typically low-dose aspirin) long term.

Perioperative management of antiplatelet therapy

- The 2024 AHA/ACC guideline for perioperative cardiovascular management for noncardiac surgery recommends strategies of deferring elective surgery ≥ 6 mo after PCI for CCS* and 12 mo after PCI for ACS.^{†‡} After this time, P2Y₁₂ inhibitors can be stopped but low-dose aspirin should be continued. In the current era, shorter DAPT durations can be considered in carefully selected patients.
- The recommended timing of stopping P2Y₁₂ inhibitors before surgery is based on the half-life of these drugs. Clopidogrel and ticagrelor are stopped 5 d before surgery and prasugrel 7 d before surgery.
- Urgent noncardiac surgery may require cessation of P2Y₁₂ agents and bridging with a shorter-acting intravenous antiplatelet agent, or the use of platelet transfusion for more urgent cases (eg, intracranial bleeding).

Monitoring of patients on antiplatelet therapy

- Considerable variability exists in patients' response to antiplatelet therapy, particularly with clopidogrel. Although routine platelet function testing and genotyping are not universally recommended, they may have a role in guiding deescalation of P2Y₁₂ inhibitor therapy.
- Routine PPI therapy is recommended for patients at high risk of bleeding or for those requiring concomitant anticoagulant use. Concerns that concomitant omeprazole reduces clopidogrel efficacy have not been proved clinically.

Adherence to antiplatelet therapy

- Medication adherence to DAPT and antiplatelet therapy remains a major challenge after MI, with nonadherence, particularly early post-PCI, associated with higher risk of stent thrombosis and MACE.
- Factors contributing to poor adherence include older age, polypharmacy, socioeconomic barriers, high drug costs, and medication-related adverse effects such as bleeding or dyspnea.
- Strategies to improve adherence may include early postdischarge follow-up, culturally sensitive patient education, addressing social drivers of health such as financial burden, close monitoring for bleeding and other adverse effects, dose adjustment or deescalation of therapy when appropriate, and potentially the use of fixed-dose polypill combinations. Multifaceted and multilevel (eg, patient, family, healthcare system) approaches are essential.

*ACC/AHA Class 2a recommendation.

†ACC/AHA Class 1 recommendation.

ACC = American College of Cardiology; ACS = acute coronary syndrome; AHA = American Heart Association; ARC-HBR = Academic Research Consortium for High Bleeding Risk; ASCVD = atherosclerotic cardiovascular disease; CABG = coronary artery bypass graft; CAC = coronary artery calcium; CAD = coronary artery disease; CCD = chronic coronary disease; CCS = chronic coronary syndrome; DAPT = dual antiplatelet therapy; DOAC = direct oral anticoagulants; HBR = high bleeding risk; MACE = major adverse cardiac events; MALE = major adverse limb events; MI = myocardial infarction; NIHSS = National Institutes of Health Stroke Scale; OAC = oral anticoagulation; PAD = peripheral artery disease; PCI = percutaneous coronary intervention; PPI = proton pump inhibitor; PRECISE-DAPT = Predicting Bleeding Complications in Patients Undergoing Stent Implantation and Subsequent Dual Antiplatelet Therapy; STEMI = ST-segment elevation MI; TIA = transient ischemic attack.

High Bleeding Risk (HBR): A system developed by the BARC to assess patients undergoing percutaneous coronary intervention (PCI) who are at risk of significant bleeding complications within 1 year after PCI.

Major adverse cardiac events (MACE): Typically defined as cardiovascular death, myocardial infarction (MI), or stroke. The exact definition can include all-cause death or cardiovascular death, but the 3-point MACE is the most commonly used endpoint for clinical trials.

MACE may also include urgent revascularization (4-point MACE) and hospitalization for unstable angina (5-point MACE).

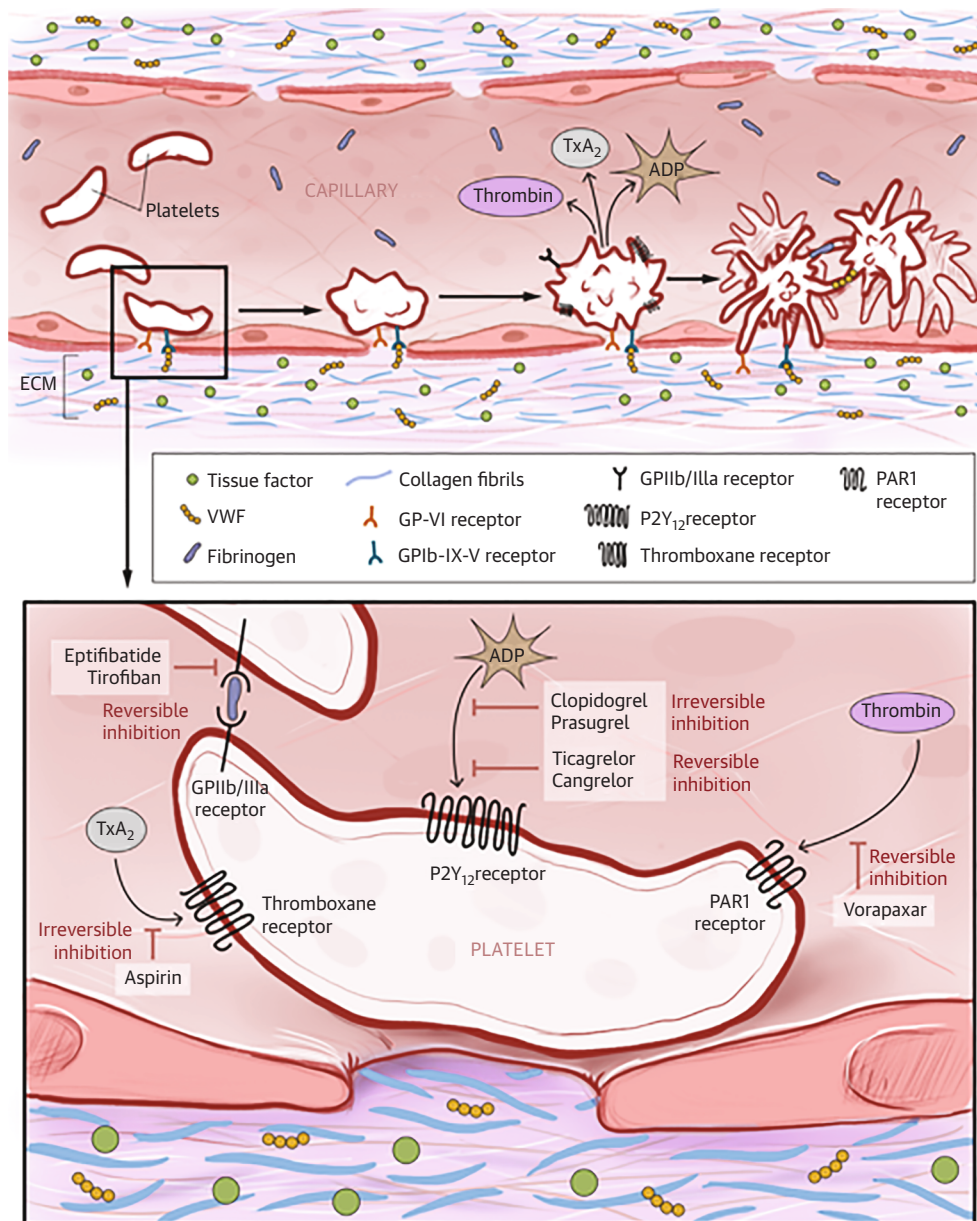
ABBREVIATIONS

Abbreviation	Meaning/Phrase
ACS	acute coronary syndrome
ADP	adenosine diphosphate
AF	atrial fibrillation
ASCVD	atherosclerotic cardiovascular disease
BARC	Bleeding Academic Research Consortium
CABG	coronary artery bypass graft
CAC	coronary artery calcium
CAD	coronary artery disease
CCD	chronic coronary disease
CCS	chronic coronary syndromes
DAPT	dual antiplatelet therapy
DES	drug-eluting stent
GPI	glycoprotein IIa/IIIb inhibitor
HBR	high bleeding risk
ICH	intracranial hemorrhage
PAD	peripheral artery disease
PCI	percutaneous coronary intervention
MACE	major adverse cardiac events
MI	myocardial infarction
PPI	proton pump inhibitor
ST	stent thrombosis
TIA	transient ischemic attack

BACKGROUND

ASCVD, which includes coronary artery disease (CAD), cerebrovascular disease, and peripheral artery disease (PAD), is the leading cause of death in the United States and worldwide. In the United States, an estimated 20.5 million adults have CAD, 7.8 million have had a stroke, and 8.5 million adults have PAD, with ~ 1.6 million MIs or strokes occurring annually.^{5,6} Platelets, a mediator of endothelial, thrombotic, immune, and inflammatory responses, are pivotal in the pathogenesis of atherothrombosis and are also involved in the initiation and progression of atherosclerosis.^{7,8} Antiplatelet agents, including but not limited to aspirin, P2Y₁₂ inhibitors, and glycoprotein IIb/IIIa inhibitors (GPI), are therefore a cornerstone in the management of ASCVD. Several newer agents are also currently being investigated.

FIGURE 1 Pharmacologic Actions of Commonly Used Antiplatelet Agents



ADP = adenosine diphosphate; ECM = extracellular matrix; GP = glycoprotein; PAR1 = protease-activated receptor 1; TxA₂ = thromboxane A₂; VWF = von Willebrand factor.

Pharmacology of commonly used antiplatelet agents

Aspirin, or acetylsalicylic acid, is the most widely used cost-effective drug to prevent platelet aggregation. It exerts its antiplatelet action mainly by irreversibly acetylating a serine residue of platelet cyclo-oxygenase (COX)-1, thus inhibiting the formation of thromboxane A₂, a potent platelet stimulator.⁹ P2Y₁₂ inhibitors (ie,

clopidogrel, prasugrel, cangrelor, ticagrelor) exert antiplatelet effects by blocking the P2Y₁₂ receptor, thereby inhibiting platelet activation through adenosine diphosphate (ADP), and limiting ADP-mediated conversion of the GP IIb/IIIa receptor to its active form (Figure 1).¹⁰⁻¹²

Among these agents, clopidogrel and prasugrel are orally administered thienopyridine prodrugs that are

converted in vivo to active metabolites and irreversibly inhibit the P2Y₁₂ receptor; however, the more potent prasugrel provides more complete, consistent platelet inhibition than clopidogrel.^{13,14} Cangrelor is an intravenous non-thienopyridine P2Y₁₂ inhibitor characterized by rapid onset (<5 minutes) and return of platelet function after cessation (≤1 hour). Ticagrelor is a potent, orally active non-thienopyridine P2Y₁₂ inhibitor that binds the receptor reversibly without requiring metabolic activation.^{15,16} GPIs such as eptifibatide and tirofiban are intravenously administered and directly block the platelet glycoprotein IIb/IIIa receptor, rapidly inhibiting fibrinogen and von Willebrand factor binding and producing near-immediate platelet inhibition within minutes. Eptifibatide and tirofiban have relatively short half-lives (~2 hours) with recovery of platelet function within 4 to 8 hours after discontinuation.¹⁴ Because aspirin and P2Y₁₂ inhibitors act via distinct mechanisms, their combination provides greater platelet inhibition than either agent alone and is referred to as dual antiplatelet therapy (DAPT). Cangrelor and GPIs have a role in the management of patients undergoing PCI, particularly those presenting with acute coronary syndromes (ACS), but are not routinely used outside of this setting.¹³ Vorapaxar is a selective antagonist of platelet protease-activated receptor-1 receptors that blocks thrombin-mediated platelet activation, one of the most potent platelet activation pathways. Although beneficial for ischemic outcomes when used along with DAPT,¹⁷ it significantly increases major bleeding and is therefore rarely used in clinical practice.

Evolution of DAPT regimens

The duration and intensity of DAPT have primarily been studied in patients undergoing PCI, presenting with ACS, or both. Due to a significantly elevated risk of stent thrombosis (ST) with first-generation drug-eluting stents (DES) (sirolimus- and paclitaxel-eluting stents) as compared with bare-metal stents,¹⁴ earlier studies on DAPT duration focused on prolonged extension, particularly because clopidogrel was the primary P2Y₁₂ inhibitor used at that time. With advances in stent technology, increased use of intravascular imaging to guide PCI, and routine use of more potent agents, attention has shifted toward mitigating bleeding risk, prompting studies of shorter DAPT duration, followed by aspirin monotherapy or deescalation to P2Y₁₂ inhibitor monotherapy after a short duration of DAPT use.⁴ Further, contemporary DES designs using ultrathin stent struts and biodegradable polymers are associated with substantially lower rates of ST, with observed rates as low as 0.4% to 0.5% at 1 year and <1% at 5 years.¹⁸

SUMMARY OF CURRENT CLINICAL PRACTICE GUIDELINE RECOMMENDATIONS

Several clinical practice guidelines provide a framework for managing patients with antiplatelet agents across the spectrum of ASCVD: 2025 ACC/American Heart Association (AHA) guideline for the management of patients with ACS,¹⁹ 2023 AHA/ACC guideline for management of patients with chronic coronary disease (CCD),²⁰ 2023 European Society of Cardiology (ESC) guidelines for the management of ACS,²¹ and 2024 ESC guidelines for the management of chronic coronary syndromes (CCS).²² A summary and comparison of recommendations from these guidelines is found in **Table 2**. Overall, recommendations regarding DAPT duration for the first year after an ACS are grouped into 3 categories: a default DAPT strategy; bleeding reduction strategies; and options to treat patients at high risk for bleeding. Twelve months of DAPT remains the default strategy and is recommended after an ACS to reduce MACE (ACC/AHA Class 1; ESC Class 1). In medically treated patients with CCD, or patients with remote coronary revascularization, aspirin is the default strategy (ACC/AHA Class 1; ESC Class 1).²⁰ In patients with CCD undergoing PCI, aspirin plus clopidogrel is recommended for 6 months as the default (ACC/AHA Class 1; ESC Class 1), or 1 to 3 months of DAPT followed by P2Y₁₂ inhibitor monotherapy (ACC/AHA Class 2a; ESC Class 2a) in certain scenarios. Although overall agreement between ACC/AHA and ESC guidelines has been determined, important differences exist. For example, the 2023 ESC ACS guidelines recommend prasugrel in preference to ticagrelor for patients with ACS who proceed to PCI (ESC Class IIa). No analogous recommendation exists in the 2025 ACC/AHA ACS guidelines.²³

ISCHEMIC VERSUS BLEEDING EVENT RISK IN DETERMINING ANTIPLATELET STRATEGY

Any ischemic benefit achieved by increasing potency of existing antiplatelet therapy is almost inevitably offset by a concomitant increase in bleeding. The optimal balance between these requires consideration of clinical and procedural features in individual patient scenarios (**Central Illustration**). These risks are closely linked; patients at high ischemic risk frequently have HBR. Temporal evolution of risk further influences antiplatelet strategy. Thrombotic risk is highest immediately after an acute ischemic event and decreases over the first month, whereas bleeding risk decreases more gradually.²⁴ Consideration for more intensive and potent initial antiplatelet therapy followed by deescalation as ischemic risk decreases can be a useful framework.^{25,26} Shared

TABLE 2 Comparison of Key Recommendations on Antiplatelet Therapy Across Major Clinical Practice Guidelines (2025 ACC/AHA ACS, 2023 ESC ACS, 2023 AHA/ACC CCD, 2024 ESC CCS)

Clinical Practice Guideline	Strength of Recommendation*	Recommendation Text
Default APT—All Comers		
2025 ACC/AHA ACS	1	In patients with ACS who are not at HBR, DAPT with aspirin and an oral P2Y ₁₂ inhibitor should be administered for at least 1 y to reduce MACE.
2023 ESC ACS	1	1. In all patients with ACS, a P2Y ₁₂ inhibitor is recommended in addition to aspirin, given as an initial oral LD followed by a MD for 12 mo unless HBR exists. 2. Aspirin is recommended for all patients without PCI at an initial oral LD of 150–300 mg PO or 75–250 mg IV and an MD of 75–100 mg OD for long-term treatment.
2023 ACC/AHA CCD	1	In patients with CCD and no indication for oral anticoagulant therapy, low-dose aspirin 81 mg (75–100 mg) is recommended to reduce atherosclerotic events.
2024 ESC CCS	1	1. In patients with CCS and a previous MI or remote PCI, aspirin 75–100 mg daily is recommended lifelong after an initial period of DAPT. 2. In patients without previous MI or revascularization but with evidence of significant obstructive CAD, aspirin 75–100 mg daily is recommended lifelong. 3. In patients with CCS and a previous MI or PCI, clopidogrel 75 mg daily is recommended as a safe and effective alternative to aspirin monotherapy.
Default APT—PCI		
2025 ACC/AHA ACS	1	1. In patients with STEMI managed with PCI, prasugrel or ticagrelor should be administered to reduce MACE and stent thrombosis. 2. In patients with NSTE-ACS undergoing PCI, prasugrel or ticagrelor is recommended to reduce MACE and stent thrombosis. 3. In patients with NSTE-ACS, clopidogrel is recommended to reduce MACE when prasugrel or ticagrelor are unavailable, cannot be tolerated, or are contraindicated.
	2a	In patients with ACS undergoing PCI with large thrombus burden, no reflow, or slow flow, adjunctive use of an intravenous or intracoronary glycoprotein IIb/IIIa inhibitor is reasonable to improve procedural success and reduce infarct size.
	2b	1. Among patients with ACS undergoing PCI who have not received a P2Y ₁₂ inhibitor, intravenous cangrelor may be reasonable to reduce periprocedural ischemic events. 2. In patients with NSTE-ACS planned for an invasive strategy with timing of angiography anticipated to be >24 h, upstream treatment with clopidogrel or ticagrelor may be considered to reduce MACE.
2023 ESC ACS	1	1. Prasugrel is recommended in P2Y ₁₂ -naïve patients proceeding to PCI. 2. Ticagrelor is recommended irrespective of the treatment strategy (invasive or conservative). 3. Clopidogrel is recommended when prasugrel or ticagrelor are unavailable, cannot be tolerated, or are contraindicated.
	2a	1. Prasugrel should be considered in preference to ticagrelor for patients with ACS who proceed to PCI. 2. Glycoprotein IIb/IIIa receptor antagonists should be considered if evidence of no-reflow or a thrombotic complication is observed during PCI.
	2b	In P2Y ₁₂ receptor inhibitor-naïve patients undergoing PCI, cangrelor may be considered.
	3	1. Pretreatment with a glycoprotein IIb/IIIa antagonist is not recommended. 2. Routine pretreatment with a P2Y ₁₂ receptor inhibitor in patients with NSTE-ACS in whom coronary anatomy is unknown and early invasive management (<24 h) is planned is not recommended.
2023 ACC/AHA CCD	1	In patients with CCD treated with PCI, DAPT consisting of aspirin and clopidogrel for 6 mo after PCI followed by SAPT is indicated to reduce MACE and bleeding events.
2024 ESC CCS	1	In patients with CCS and no indication for oral anticoagulation, DAPT consisting of aspirin 75–100 mg and clopidogrel 75 mg daily for up to 6 mo is recommended as the default antithrombotic strategy after PCI-stenting.
Default DAPT—Non-PCI		
2025 ACC/AHA ACS	1	1. In patients with STEMI managed with fibrinolytic therapy, clopidogrel should be administered concurrently to reduce death and MACE. 2. In patients with NSTE-ACS who are managed without planned invasive evaluation, ticagrelor is recommended to reduce MACE. 3. In patients with NSTE-ACS, clopidogrel is recommended to reduce MACE when prasugrel or ticagrelor are unavailable, cannot be tolerated, or are contraindicated.
2023 ESC ACS	1	Aspirin and clopidogrel are recommended [when combining APT with fibrinolysis].
Routine Deescalation Post-PCI		
2025 ACC/AHA ACS	1	In patients with ACS who have tolerated DAPT with ticagrelor, transition to ticagrelor monotherapy ≥1 mo after PCI is useful to reduce bleeding risk.
	2b	1. In patients with ACS undergoing PCI, deescalation of DAPT (switching from ticagrelor or prasugrel to clopidogrel) after 1 mo may be reasonable to reduce bleeding risk. 2. In patients with ACS undergoing PCI who are at HBR, transition to SAPT (aspirin or P2Y ₁₂ inhibitor) after 1 mo may be reasonable to reduce bleeding risk.

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TABLE 2 Continued

Clinical Practice Guideline	Strength of Recommendation*	Recommendation Text
2023 ESC ACS	2a	In patients who are event-free after 3–6 mo of DAPT and who are not high ischemic risk, SAPT (preferably with a P2Y ₁₂ receptor inhibitor) should be considered.
	2b	In patients with HBR, aspirin or P2Y ₁₂ receptor inhibitor monotherapy after 1 mo of DAPT may be considered
	3	Deescalation of APT in the first 30 d after an ACS event is not recommended.
2023 ACC/AHA CCD	2a	In selected patients with CCD treated with PCI and a DES who have completed a 1- to 3-mo course of DAPT, P2Y ₁₂ inhibitor monotherapy for at least 12 mo is reasonable to reduce bleeding risk.
2024 ESC CCS	2b	Stopping DAPT after 1–3 mo from PCI stenting may be considered in patients who are not at HBR nor at high risk of ischemic events.
Need for Concomitant OAC or Other High Bleeding Risk		
2025 ACC/AHA ACS	1	In patients with ACS who require oral anticoagulant therapy, aspirin should be discontinued after 1–4 wks of triple antithrombotic therapy, with continued use of a P2Y ₁₂ inhibitor (preferably clopidogrel) and an oral anticoagulant to reduce bleeding risk.
2023 ESC ACS	1	As the default strategy for patients with atrial fibrillation and CHA ₂ DS ₂ -VASC score ≥1 in men and ≥2 in women, after up to 1 wk of triple antithrombotic therapy after the ACS event, dual antithrombotic therapy using a non-VKA OAC at the recommended dose for stroke prevention and a single oral antiplatelet agent (preferably clopidogrel) for up to 12 mo is recommended.
	2a	In patients treated with an OAC, aspirin plus clopidogrel for >1 wk and up to 1 mo should be considered in those with high ischemic risk or with other anatomic-procedural characteristics that are judged to outweigh the bleeding risk.
	2b	In older patients with ACS, especially if HBR, clopidogrel as the P2Y ₁₂ receptor inhibitor may be considered.
	3	The use of ticagrelor or prasugrel as part of triple APT is not recommended.
2023 ACC/AHA CCD	1	In patients with CCD who have undergone elective PCI and who require OAC therapy, DAPT for 1 to 4 wks followed by clopidogrel alone for 6 mo should be administered in addition to direct OAC.
	2a	In patients with CCD who have undergone PCI and who require OAC therapy, continuing aspirin in addition to clopidogrel for up to 1 mo is reasonable if the patient has a high thrombotic risk and low bleeding risk.
	2b	1. In patients with CCD who require OAC and have a low atherothrombotic risk, discontinuation of aspirin therapy with continuation of direct OAC alone may be considered 1 y after PCI to reduce bleeding risk. 2. In patients with CCD who require oral anticoagulation, direct OAC monotherapy may be considered if there is no acute indication for concomitant APT.
2024 ESC CCS	1	1. In patients at HBR but not at high ischemic risk, it is recommended to discontinue DAPT 1–3 mo after PCI and to continue with SAPT. 2. In patients with an indication for OAC who undergo PCI, initial low-dose aspirin once daily is recommended (LD when not on MD) in addition to OAC and clopidogrel. 3. After uncomplicated PCI in patients with CCS and concomitant indication for OAC: • Early cessation of aspirin (≤1 wk); • Followed by continuation of OAC and clopidogrel: ○ Up to 6 mo in patients not at high ischemic risk OR ○ Up to 12 mo in patients at high ischemic risk; • Followed by OAC alone; is recommended. 4. In patients with CCS and a long-term indication for OAC, an AF therapeutic dose of VKA alone or, preferably, of DOAC alone (unless contraindicated) is recommended lifelong. 5. In patients who are eligible for OAC, DOAC (unless contraindicated) is recommended in preference to VKA.
	2a	1. Continuation of aspirin up to 1 mo after PCI, in addition to OAC and clopidogrel, should be considered in patients at high ischemic risk or with anatomic-procedural characteristics judged to outweigh the bleeding risk. 2. When concerns about HBR prevail over concerns about stent thrombosis or ischemic stroke, rivaroxaban 15 mg daily should be considered in preference to rivaroxaban 20 mg daily for the duration of concomitant APT. 3. Dabigatran 110 mg twice daily should be considered in preference to dabigatran 150 mg twice daily for the duration of concomitant APT. 4. In patients with an indication for VKA in combination with single or dual APT, targeting VKA intensity to an INR in the lower part of the recommended range and to a time in therapeutic range >70% should be considered
	2b	In patients requiring OAC, withdrawing APT at 6 mo while continuing OAC may be considered.
Long-Term APT		
2023 ACC/AHA CCD	2a	In patients with CCD without an indication for therapeutic DOAC or DAPT and who are at high risk of recurrent ischemic events but low-to-moderate bleeding risk, the addition of low-dose rivaroxaban 2.5 mg twice daily to aspirin 81 mg daily is reasonable for long-term reduction of risk for MACE.
	2b	1. In patients with CCD who have had a previous MI and are at low bleeding risk, extended DAPT beyond 12 mo for a period of up to 3 y may be reasonable to reduce MACE. 2. In patients with CCD and a previous history of MI without a history of stroke, TIA, or ICH, vorapaxar may be added to aspirin therapy to reduce MACE.

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TABLE 2 Continued

Clinical Practice Guideline	Strength of Recommendation*	Recommendation Text
2024 ESC CCS	2a	Adding a second antithrombotic agent to aspirin for extended long-term secondary prevention should be considered in patients at enhanced ischemic risk and without high bleeding risk.
	2b	In patients with CCS undergoing high-thrombotic risk stenting (eg, complex left main stem, 2-stent bifurcation, suboptimal stenting result, previous stent thrombosis [previously known CYP2C19 *2/*3 polymorphisms]), prasugrel or ticagrelor (in addition to aspirin) may be considered instead of clopidogrel, for the first month, and up to 3–6 mo.
Post-CABG		
2025 ACC/AHA ACS	1	Continue aspirin during CABG and start P2Y ₁₂ inhibitor when safe postoperatively.
2023 ESC ACS	1	If patients presenting with ACS stop DAPT to undergo CABG, it is recommended that they resume DAPT after surgery for at least 12 mo.
2023 ACC/AHA CCD	2b	In patients with CCD, the use of DAPT after CABG may be useful to reduce the incidence of saphenous vein graft occlusion.
2024 ESC CCS	1	1. It is recommended to initiate aspirin postoperatively as soon as there is no concern over bleeding (post-CABG). 2. After CABG, aspirin 75–100 mg daily is recommended lifelong.
	2b	DAPT may be considered after CABG in selected patients at greater risk of graft occlusion and at low risk of bleeding.
Use of PPI		
2025 ACC/AHA ACS	1	In patients at high risk of gastrointestinal bleeding, a PPI is recommended in combination with DAPT, oral anticoagulants, or both to reduce risk of bleeding.
2023 ESC ACS	1	A PPI in combination with DAPT is recommended in patients at high risk of gastrointestinal bleeding.
2023 ACC/AHA CCD	2a	In patients with CCD on DAPT, the use of a PPI can be effective in reducing gastrointestinal bleeding risk.

*ACC/AHA Class 1 or ESC Class 1 = recommended/indicated based on evidence; ACC/AHA Class 2a or ESC Class 2a = weight of evidence favors efficacy; ACC/AHA Class 2b or ESC Class 2b = weight of evidence is less well established; ACC/AHA Class 3 or ESC Class 3 = not recommended based on evidence of harm.

ACC = American College of Cardiology; ACS = acute coronary syndromes; AF = atrial fibrillation; AHA = American Heart Association; APT = antiplatelet therapy; CAD = coronary artery disease; CCD = chronic coronary disease; CCS = chronic coronary syndromes; DAPT = dual antiplatelet therapy; DES = drug-eluting stent; ESC = European Society of Cardiology; HBR = high bleeding risk; ICH = intracranial hemorrhage; INR = international normalized ratio; LD = loading dose; MACE = major adverse cardiac events; MD = maintenance dose; MI = myocardial infarction; OAC = oral anticoagulant; PCI = percutaneous coronary intervention; NSTE-ACS = non-ST-elevation acute coronary syndrome; SAPT = single antiplatelet therapy; STEMI = ST-elevation myocardial infarction; VKA = vitamin K antagonist.

decision-making is an important aspect of individualized risk-benefit discussions among patients with ASCVD.

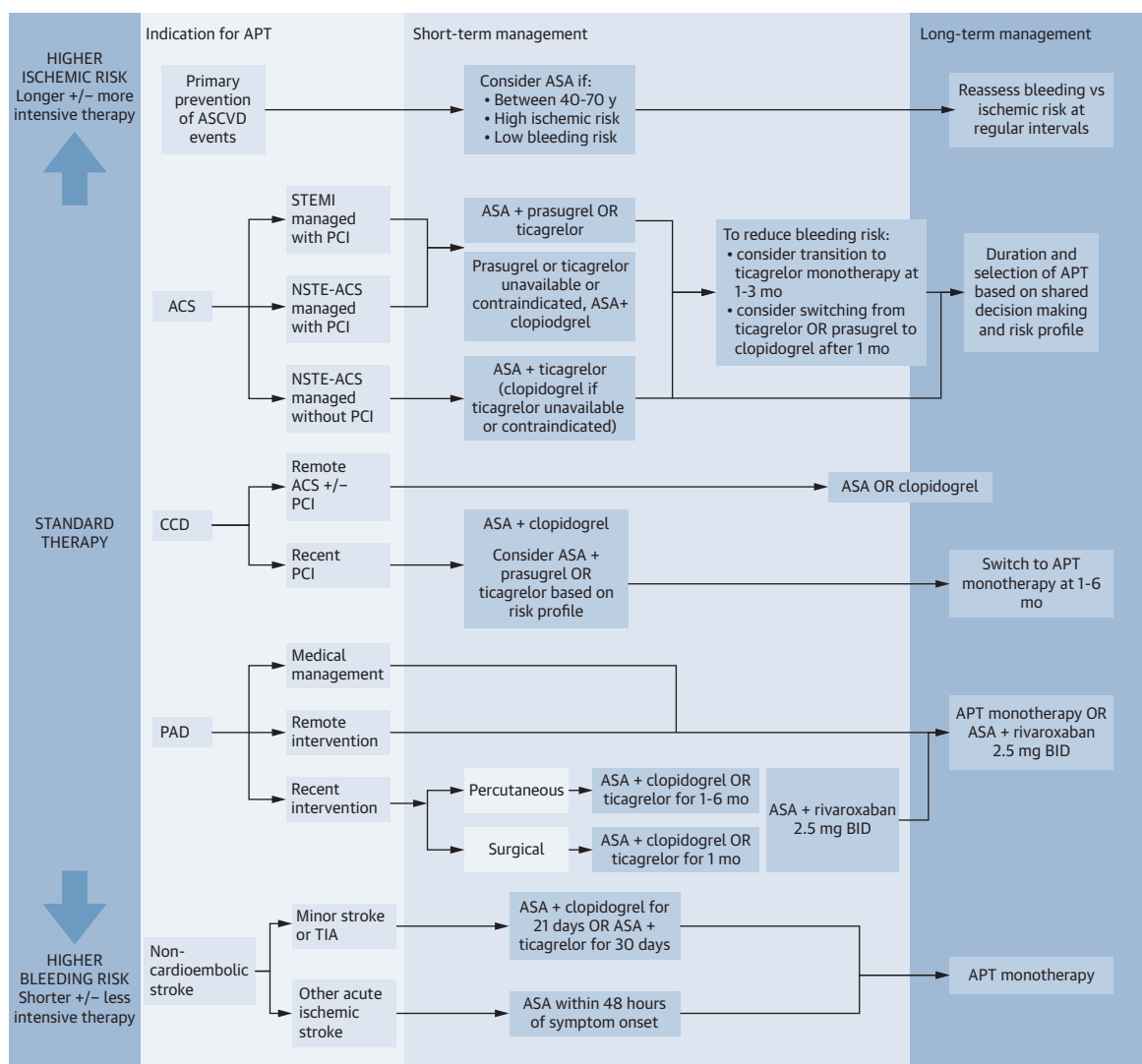
Certain conditions are associated with higher future ischemic risk. These include ACS,¹⁹ complex PCI for stable CAD,²⁰ recent stroke,²⁸ or recent acute limb ischemic event.²⁹ Clinical characteristics that predict bleeding include advanced age, previous stroke or transient ischemic attack (TIA), renal dysfunction, low body weight, female sex, anemia, and previous bleeding. The widely used Academic Research Consortium for High Bleeding Risk (ARC-HBR) score integrates ≥ 1 major or ≥ 2 minor criteria, associated with a 1-year risk of BARC Type 3 to 5 bleeding $\geq 4\%$ or intracranial hemorrhage risk $\geq 1\%$.^{30,31} The ARC-HBR was designed to standardize characterizing major bleeding risk for individuals undergoing PCI.

Antiplatelet therapy carries a low overall incidence of intracranial hemorrhage (ICH), one of the most serious complications, typically estimated at 2 to 3 cases per 1,000 patient years.³⁰ However, morbidity and mortality of a major bleeding event can equal or exceed that of an ischemic event.³⁰ Rates of ICH, fatal bleeding, and nonfatal bleeding are increased among patients treated with DAPT compared with aspirin monotherapy.³² Despite this, no difference in all-cause death was

observed, and lower rates of ischemic events such as MI and stroke seen with DAPT occurred.³²

Among patients at higher risk of recurrent ischemic events, ischemic and bleeding risks often increase in parallel due to shared risk factors; however, the absolute risk of ischemic events typically exceeds that of major bleeding. In patients with previous noncardioembolic stroke or TIA, annual ischemic event rates range from 2.5% to 10.2% across risk strata, compared with major bleeding rates of 0.7% to 3.2% on aspirin monotherapy.³³ Similar patterns are observed in CAD: in a pooled cohort of patients with previous MI, the hazard ratio (HR) for death was 6.1 after ischemic events, compared with 3.7 for BARC Type 3 to 5 bleeding, relative to patients without either event. When death weighting is used to assess the clinical impact, 44% of patients considered HBR had a greater risk of an ischemic event than a significant bleeding event, and 32% had comparable risk of both, highlighting challenge of decision-making focused on bleeding risk.³⁰ Risk stratification tools are commonly used in clinical decision-making to help balance ischemic and bleeding risks (Table 3); however, important limitations are noted. Dual Antiplatelet Risk Score³⁴ and PRECISE-DAPT (Predicting Bleeding Complications in

CENTRAL ILLUSTRATION Indications for Antiplatelet Use by Clinical Scenario



Kumbhani DJ, et al. JACC. 2026;88(11):1284-1313.

Adapted with permission from Kumbhani et al.²⁷ ACS = acute coronary syndrome; APT = antiplatelet therapy; ASA = low dose aspirin (75-100 mg); ASCVD = atherosclerotic cardiovascular disease; CCD = chronic coronary disease; DAPT = dual antiplatelet therapy; mo = months; NSTEMI = non-ST-elevation ACS; PAD = peripheral artery disease; PCI = percutaneous coronary intervention; STEMI = ST-elevation myocardial infarction; TIA = transient ischemic attack.

Patients Undergoing Stent Implantation and Subsequent Dual Antiplatelet Therapy³⁵ are used to guide DAPT duration after PCI. In patients with stroke or TIA, the S₂TOP-BLEED (Sex, Smoking, Type of antiplatelet, Outcome on modified Rankin Scale Score, Prior stroke, Blood pressure, Low BMI, Elderly, Ethnicity, Diabetes) score estimates major bleeding and ICH risk with the use of antiplatelets based on independent risk factors for bleeding.³³ These calculators are summarized in **Table 3** and can aid point-of-care discussions with

patients. No single clinical calculator is inclusive of all variables associated with higher bleeding, and scores must be contextualized for an individual patient.

Consensus recommendations: ischemic vs bleeding risk in determining antiplatelet strategy

- Shared decision-making for use of antiplatelet therapies is centered on understanding the risk of patient-specific future ischemic events compared with bleeding risk.

TABLE 3 Comparison of Commonly Used Bleeding Risk Calculators

	DAPT	PRECISE-DAPT	ARC-HBR	S ₂ TOP-BLEED
When to use	12 mo post-PCI	Immediately after PCI	Post-PCI to identify patients at high risk of major bleeding or ICH	After noncardioembolic stroke or TIA
Output	Balance of bleeding vs ischemic risk for longer-term DAPT compared with aspirin monotherapy	Baseline bleeding risk to guide duration of DAPT up to 12 mo	Balance of bleeding vs ischemic risk post-PCI	Risk of major bleeding with use of antiplatelets
Strata	■ Score ≥ 2: Favorable benefit-risk ratio for prolonged DAPT ■ Score < 2: Greater bleeding risk without increased benefit to reduce ischemic events	■ Score ≥ 25: Shorter DAPT durations (3–6 mo) may be preferred if ischemic risk is low due to higher bleeding risk ■ Score < 25: Standard or extended DAPT may be reasonable, especially if ischemic risk is high	■ HBR: At least 1 major or 2 minor criteria are met; shorter DAPT durations (3–6 mo) may be preferred if ischemic risk is low due to higher bleeding risk ■ Non-HBR: Above criteria not met	■ Score 0–10: Low bleeding risk ■ S₂TOP-BLEED score 11–15: medium bleeding risk ■ Score ≥ 16: High bleeding risk; shorter DAPT durations (3–6 mo) may be preferred if ischemic risk is low due to higher bleeding risk
Factors considered	■ Age ■ Cigarette smoking ■ Diabetes mellitus ■ MI at presentation ■ Previous PCI or MI ■ Paclitaxel-eluting stent ■ Stent diameter < 3 mm ■ CHF or LVEF $< 30\%$ ■ Vein graft stent	■ Hemoglobin ■ Age ■ WBC ■ CrCl ■ Previous bleeding	Major criteria: ■ Anticipated use of long-term oral anticoagulation ■ Severe or end-stage CKD (eGFR < 30 mL/min) ■ Hemoglobin < 11 g/dL ■ Spontaneous bleeding requiring hospitalization or transfusion in the past 6 mo or at any time, if recurrent ■ Moderate or severe baseline thrombocytopenia (platelet count $< 100 \times 10^9/L$) ■ Chronic bleeding diathesis ■ Liver cirrhosis with portal hypertension ■ Active malignancy (excluding non-melanoma skin cancer) within the past 12 mo ■ Previous spontaneous ICH (at any time) ■ Previous traumatic ICH within the past 12 mo ■ Presence of a brain arteriovenous malformation ■ Moderate or severe ischemic stroke within the past 6 mo ■ Nondeferrable major surgery on DAPT ■ Recent major surgery or major trauma within 30 d before PCI Minor: ■ Age ≥ 75 y ■ Moderate CKD (eGFR 30–59 mL/min) ■ Hemoglobin 11–12.9 g/dL for men and 11–11.9 g/dL for women ■ Spontaneous bleeding requiring hospitalization or transfusion within the past 12 mo not meeting the major criterion ■ Long-term use of oral NSAIDs or steroids ■ Any ischemic stroke at any time not meeting the major criterion	■ Sex ■ Smoking ■ Type of antiplatelet ■ Outcome (mRS) ■ Previous stroke ■ Blood pressure ■ Lower BMI ■ Age ■ Ethnicity ■ Diabetes mellitus

ARC-HBR = Academic Research Consortium for High Bleeding Risk; BMI = body mass index; CHF = congestive heart failure; CKD = chronic kidney disease; CrCl = creatinine clearance; DAPT = dual antiplatelet therapy; eGFR = estimated glomerular filtration rate; HBR = high bleeding risk; Hgb = hemoglobin; ICH = intracranial hemorrhage; LVEF = left ventricular ejection fraction; MI = myocardial infarction; mRS = modified Rankin scale; NSAID = nonsteroidal anti-inflammatory drug; PCI = percutaneous coronary intervention; PRECISE-DAPT = Predicting Bleeding Complications in Patients Undergoing Stent Implantation and Subsequent Dual Antiplatelet Therapy; TIA = transient ischemic attack; WBC = white blood cells.

- Using risk calculators such as DAPT, PRECISE-DAPT, and ARC-HBR can be helpful, but decision-making requires global assessment of future ischemic risk balanced against bleeding risk.

ANTIPLATELET THERAPY IN PRIMARY PREVENTION OF MACE

A large body of evidence demonstrates that prophylactic low-dose aspirin reduces the incidence of MI and stroke,

but not death. Because aspirin increases major bleeding risk, its use for primary prevention is limited to individuals at high cardiovascular risk, avoiding routine use in adults ≥ 70 years of age where the risk-benefit balance is unfavorable. Commonly used risk calculators include the PCE (Pooled Cohort Equations)³⁶ and PREVENT (Predicting Risk of Cardiovascular Disease EVENTSs).³⁷

Efficacy and safety of aspirin in primary prevention of MACE have been studied in a meta-analysis of 13 large-

scale trials including 165,265 participants with 1,050,511 participant-years showing a modest reduction in cardiovascular outcomes (HR, 0.89), but at an increase of major bleeding (HR, 1.43).³⁸ Interestingly, all-cause death was unchanged and the protective effects were only seen for MI and stroke. Excess gastrointestinal bleeding (HR, 1.56) was the most common major bleeding, followed by excess intracranial bleeding (HR, 1.34). Although the incidence of the latter was rare, the number needed to harm with aspirin is 880 for this disastrous complication; the number needed to treat in this analysis was 241. Another meta-analysis using much of the same trial data found similar benefit and risk of antiplatelet therapy for primary prevention.³⁹ Interestingly, this analysis showed that the benefit of antiplatelet therapy was highest among trials started before 2000 (relative risk, 0.78; a significant reduction of MI of 22%) and seemed to decrease among studies started thereafter (relative risk, 0.90; a nonsignificant reduction of MI of 10%). It may be that the introduction of modern background preventive strategies has played a role. Contemporary lipid-lowering therapy was rare in the earlier trials (0%-16%), but common in the most recent ones (up to 75%).⁴⁰ Unfortunately, head-to-head comparisons between a statin versus aspirin are lacking. It is likely that they have a synergistic effect in primary prevention as shown in TIPS-3 (The International Polycap Study 3) trial, a study of aspirin alone versus a polypill containing simvastatin and aspirin.⁴¹ In secondary prevention, this synergism has been shown in randomized trials with pravastatin.⁴²

Coronary artery calcium (CAC) scoring has emerged as a powerful tool for further risk stratifying patients who have subclinical atherosclerosis and may therefore benefit from intensified risk reduction before experiencing MACE. Analyses of data from MESA (Multiethnic Study of Atherosclerosis) and DHS (Dallas Heart Study) found that a CAC score of ≥ 100 and particularly ≥ 400 could be useful for individuals in whom the benefit-harm ratio is likely to be favorable for aspirin preventing MACE.^{43,44} Similarly, measurement of lipoprotein (a) can refine estimations of cardiac risk,⁴⁵⁻⁴⁷ but the benefit of aspirin as a preventive strategy in those with elevated lipoprotein (a) appears limited to older carriers of specific genotypes.⁴⁸ Randomized trials on antiplatelet therapy as primary prevention in those with elevated CAC level, lipoprotein (a) level, or both are currently lacking.

Consensus recommendations: aspirin for primary prevention of ASCVD events

- Aspirin may be effective in primary prevention of cardiovascular events, but is associated with increased risk of major bleeding.¹ The current lack of benefit with aspirin for primary prevention may be due to the

application of other effective preventive strategies like statins.

- Low-dose aspirin might be considered for primary prevention of ASCVD events among adults aged 40-70 years who have a higher cardiovascular risk without an increased bleeding risk.¹
- Routine use of aspirin for prevention in those over age 70 years should be avoided.
- Available ASCVD calculators may help identify patients at high ischemic risk. If these patients are at low bleeding risk, aspirin may be considered for primary prevention. Use of aspirin in those with HBR is associated with net harm regardless of CAC score or ASCVD risk.

SECONDARY PREVENTION: ACUTE MANAGEMENT

Choice of P2Y₁₂ in ACS

Acute coronary presentations of ST-elevation or non-ST-elevation MI impose a high risk of reinfarction and other adverse ischemic events particularly within the first month. Among patients undergoing PCI, the PLATO (PLATElet inhibition and patient Outcomes)⁴⁹ and the TRITON-TIMI 38 (Trial to Assess Improvement in Therapeutic Outcomes by Optimizing Platelet Inhibition with Prasugrel Thrombosis in MI 38)⁵⁰ studies provide the basis for the recommended strategy of DAPT with low-dose aspirin and a potent P2Y₁₂ inhibitor (prasugrel or ticagrelor). Clopidogrel should be used with aspirin only in patients with contraindications to more potent agents, those with HBR, or in situations where these agents are unavailable or unaffordable. Because prasugrel did not demonstrate superiority over clopidogrel in medically managed patients with ACS in the TRILOGY-ACS (Targeted Platelet Inhibition to Clarify the Optimal Strategy to Medically Manage Acute Coronary Syndromes) trial, treatment options in this population include clopidogrel or ticagrelor, with a preference for ticagrelor over clopidogrel unless unavailable or contraindicated.⁵¹ In patients with ACS treated with PCI, the ISAR-REACT (Intracoronary Stenting and Antithrombotic Regimen Rapid Early Action for Coronary Treatment)-5 study showed that DAPT with prasugrel led to lower risks of death, MI, or stroke without increased bleeding compared with DAPT with ticagrelor.⁵² Similar results were noted in the TUXEDO-2 (The Ultrathin Strut vs Xience in a Diabetic Population With Multivessel Disease 2) trial, in which 80% were enrolled with ACS.⁵³ For patients without contraindication, choosing prasugrel over ticagrelor is reasonable and aligns with the 2024 ESC ACS guidelines (Class IIa recommendation). Additionally, once-daily dosing may offer advantages for medication adherence. The Popular Age (clopidogrel versus

ticagrelor or prasugrel in patients ≥ 70 years with non-ST-elevation ACS) study and the TRILOGY subgroup study showed that in patients ≥ 70 years of age who have higher risks of ischemia and bleeding, clopidogrel offers less bleeding risk with similar ischemic risk than DAPT with prasugrel or ticagrelor.^{54,55} For all patients undergoing PCI, aspirin should be maintained in the periprocedural period.

Antiplatelet therapy with PCI for ACS

Pretreatment with oral P2Y₁₂ inhibitors

Pretreatment with oral P2Y₁₂ inhibitors may be considered for ST-elevation MI but is usually not recommended for non-ST-elevation MI until the coronary anatomy is known because this either shows no benefit or a higher bleeding risk.^{52,56} Pretreatment may be considered for non-ST-elevation ACS when an invasive treatment strategy is planned but angiography will be delayed by >24 hours, although times to angiography in the current era for this patient population are relatively short.⁵⁷

Intravenous antiplatelet drugs during PCI

Intravenous antiplatelet drugs include GPIs (eptifibatide and tirofiban) and cangrelor. GPIs were assessed before routine use of potent oral P2Y₁₂ inhibitors (ticagrelor and prasugrel). They increase the risk of bleeding after PCI, especially with femoral artery access, and are generally reserved for bailout for thrombotic complications or no-reflow after PCI. Three trials from the CHAMPION (Cangrelor versus Standard Therapy to Achieve Optimal Management of Platelet Inhibition) group of studies compared cangrelor to placebo followed by DAPT with clopidogrel.⁵⁸⁻⁶⁰ These trials reported a small but significant decrease in ischemic events with cangrelor compared with clopidogrel, with an incremental increase in bleeding.⁶¹ Of note, no trials directly compare cangrelor use to prasugrel or ticagrelor loading.⁶²

Duration of DAPT after PCI

The default duration of DAPT for 12 months has been increasingly questioned in the era of newer-generation DES, particularly in patients at HBR. Several trials evaluated abbreviated DAPT followed by P2Y₁₂ inhibitor monotherapy (Table 4). In the TWILIGHT (Ticagrelor with Aspirin or Alone in High-Risk Patients after Coronary Intervention) trial,⁶³ high-risk patients who completed 3 months of uneventful DAPT with ticagrelor plus aspirin were randomized to continue DAPT or switch to ticagrelor monotherapy. Discontinuation of aspirin significantly reduced BARC Types 2, 3, or 5 bleeding without increasing death, MI, or stroke.⁶⁴ Similar findings were observed in the TICO (Ticagrelor Monotherapy After 3 Months), T-PASS (Ticagrelor Monotherapy in Patients Treated With New-Generation DES for ACS), and

TARGET-FIRST (Evaluation of a Modified Anti-Platelet Therapy Associated With Low-dose DES Firehawk in Acute MI Patients Treated With Complete Revascularization Strategy) trials, where 1 to 3 months of DAPT followed by ticagrelor or prasugrel monotherapy reduced major bleeding with comparable ischemic outcomes versus 12 months of DAPT.^{57,64,65} Most of these trials studied ticagrelor monotherapy; data with prasugrel are less robust. By contrast, in STOPDAPT-2 ACS (Short and Optimal Duration of Dual AntiPlatelet Therapy-2 Study),⁶⁶ 1 to 2 months of DAPT followed by clopidogrel monotherapy was not found to be noninferior to a strategy of standard 12 months of DAPT in the primary composite ischemic and bleeding endpoint and was associated with a strong trend toward increased ischemic events. Therefore, abbreviated DAPT followed by clopidogrel monotherapy is not routinely used in patients with ACS undergoing PCI. Notably, most of these trials were powered to assess bleeding outcomes rather than ischemic endpoints. Nevertheless, they did not demonstrate an increase in ischemic events among patients randomized to P2Y₁₂ monotherapy. Importantly, trials examining a shortened duration of DAPT specified discontinuation of aspirin after 1 to 3 months. In the NEOMINDSET (Percutaneous Coronary Intervention Followed by Antiplatelet Monotherapy in the Setting of Acute Coronary Syndromes) trial, very early deescalation by discontinuing aspirin and switching to ticagrelor or prasugrel monotherapy 4 days after PCI was associated with higher ischemic risk.⁶⁷ Aspirin monotherapy after 3 to 6 months of DAPT following PCI for ACS decreases the risk of major bleeding, but some studies suggest higher ischemic risk.⁶⁷⁻⁷⁰ Thus, monotherapy with ticagrelor or prasugrel after 1 to 3 months of DAPT after PCI for ACS is reasonable but not monotherapy with aspirin or clopidogrel unless the patient has HBR. Although meta-analyses support this,⁷¹ ischemic event rates in the individual trials were lower than expected, and P2Y₁₂ inhibitor monotherapy after shorter durations may not be generalizable to patients at higher risk of ischemia, such as those with ST-elevation MI.

Other deescalation strategies

Another approach to reducing bleeding risk is to deescalate by switching from DAPT with more potent inhibitors (ticagrelor or prasugrel) to clopidogrel.⁴ Deescalation can be guided by platelet function tests or genotyping, or unguided. Trials in which DAPT with either ticagrelor or prasugrel was deescalated by unguided switching to DAPT with clopidogrel 1 month after PCI for ACS were noninferior for ischemic outcomes to continuing the more potent DAPT for 12 months.⁷²⁻⁷⁶ A meta-analysis of these studies concludes that these strategies reduce major bleeding without increasing the

TABLE 4 Landmark Randomized Clinical Trials Evaluating Long-Term Antithrombotic Therapy for CAD

Trial	Population	Comparison	Follow-Up	Ischemic Findings	Bleeding Findings	Key Takeaways
Aspirin-Based Strategies ≥1 y After ACS						
CARDIFF 1,2 ^{90,91}	Prior MI	Aspirin vs placebo	Long term	↓ Recurrent MI, vascular death	↑ Minor bleeding	Aspirin superior to placebo
PARIS ⁹²	Prior MI	Aspirin vs placebo	Long term	↓ Vascular events	↑ Bleeding events	Aspirin superior to placebo
CDPA ⁹³	Prior MI	Aspirin vs placebo	Long term	↓ MI and stroke	↑ Minor bleeding	Supports chronic aspirin
AMIS ⁹⁴	Acute MI survivors	Aspirin vs placebo	3 y	↓ Reinfarction, death	↑ Minor bleeding	Secondary prevention benefit
GAMIS ⁹⁵	Post-MI	Aspirin vs placebo	Long term	↓ Events	↑ Bleeding	Confirms benefit
ATT meta-analyses ⁹⁶	Vascular disease	Low- vs high-dose aspirin	Variable	Similar efficacy	↓ Bleeding with low dose	Low-dose aspirin preferred
ADAPTABLE ⁹⁸	ASCVD	Low- vs high-dose aspirin	26 mo	Similar MACE	Similar major bleeding	81 mg favored (adherence)
Aspirin vs prolonged DAPT						
DAPT ⁹⁹	Post-PCI	DAPT >12 mo vs aspirin	30 mo	↓ MI, stent thrombosis	↑ Major bleeding	Benefit in selected high-risk patients
PEGASUS-TIMI 54 ¹⁰⁰	Prior MI (1–3 y)	Ticagrelor + aspirin vs aspirin	33 mo	↓ MACE	↑ Major bleeding	Net benefit only if low bleed risk
P2Y₁₂ Inhibitor vs DAPT						
GLOBAL LEADERS ⁸⁴	PCI	Ticagrelor monotherapy vs DAPT	24 mo	Neutral	No bleeding reduction	No superiority
TWILIGHT ⁶³	High-risk PCI	Ticagrelor monotherapy vs ticagrelor + aspirin after 3 mo	12 mo	Noninferior ischemia	↓ Major bleeding	Supports aspirin withdrawal
STOPDAPT-2 ¹⁰¹	PCI	DAPT vs clopidogrel monotherapy after 1 mo	12 mo	Similar ischemia	↓ Bleeding	Feasible in selected patients
TARGET-FIRST ⁵⁶	High-risk PCI	P2Y ₁₂ inhibitor monotherapy (ticagrelor 74%, prasugrel 21%, clopidogrel 5%) vs DAPT after 1 mo	12 mo	Neutral	↓ Major bleeding	Supports P2Y ₁₂ inhibitor monotherapy after 1 mo
Aspirin vs P2Y₁₂ inhibitor monotherapy						
CAPRIE ¹⁰²	Coronary and vascular disease	Clopidogrel vs aspirin	1 y	9% lower ischemic risk with clopidogrel	Similar bleeding	Clopidogrel alternative to aspirin
SMART CHOICE 3 ¹⁰³	PCI	Clopidogrel vs aspirin	2.3 y	Lower ischemic events	Similar bleeding	Clopidogrel may be superior to aspirin
HOST-EXAM ¹⁰⁴	PCI	Clopidogrel vs aspirin	10.5 y	Lower ischemic events	Lower bleeding	Clopidogrel may be superior to aspirin
TICAB ¹⁰⁵	Post-CABG	Ticagrelor vs aspirin	12 mo	Neutral	Similar bleeding	No added benefit
DACAB ¹⁰⁶	Post-CABG	Ticagrelor ± aspirin vs aspirin	12 mo	Neutral	↑ Bleeding with combo	Aspirin adequate post-CABG
Aspirin vs Oral Anticoagulation and Combination Strategies						
WARIS I ¹⁰⁷	Post-MI	Warfarin vs aspirin	3 y	Similar event reduction	↑ Bleeding	No advantage over aspirin
WARIS II ¹⁰⁸	Post-MI	Warfarin ± aspirin vs aspirin	4 y	↓ MI	↑ Major bleeding	Bleeding offsets benefit
COMPASS ¹⁰⁹	Stable CAD/PAD	Full dose rivaroxaban vs aspirin	23 mo	Similar CV ischemic events	↑ Major bleeding	No benefit for full-dose OAC alone
COMPASS ¹⁰⁹	Stable CAD/PAD	Low-dose rivaroxaban with aspirin vs aspirin	23 mo	Lower CV ischemic events	↑ Major bleeding	Net benefit for half-dose OAC with aspirin
APPRAISE 2 ¹¹⁰	Recent ACS	DOAC + DAPT vs DAPT	6 mo	No ischemic benefit	↑ Bleeding	Strategy unfavorable
ATLAS ACS 2 ¹¹¹	Recent ACS	Low-dose rivaroxaban + DAPT	13 mo	↓ CV death, MI	~3× ↑ bleeding	Net benefit uncertain
GEMINI-ACS-1 ¹¹²	Recent ACS	Rivaroxaban + P2Y ₁₂ inhibitor vs DAPT	6–12 mo	Similar ischemia	Similar bleeding	Feasible but not superior

ACS = acute coronary syndrome; ADAPTABLE = Aspirin Dosing: A Patient-Centric Trial Assessing Benefits and Long-Term Effectiveness; AMIS = Aspirin Myocardial Infarction Study; APPRAISE-2 = Apixaban Plus Mono Versus Dual Antiplatelet Therapy in Acute Coronary Syndromes; APT = antiplatelet trial; ATLAS-ACS 2 = Anti-Xa Therapy to Lower Cardiovascular Events in Addition to Standard Therapy in Subjects with Acute Coronary Syndrome; CAD = coronary artery disease; CAPRIE = Clopidogrel versus Aspirin in Patients at Risk of Ischemic Events; CDPA = Coronary Drug Project Aspirin; COMPASS = Rivaroxaban with or without Aspirin in Stable Cardiovascular Disease; CV = cardiovascular; DACAB = Different Antiplatelet Therapy Strategy After Coronary Artery Bypass Graft Surgery; DAPT = dual antiplatelet therapy; DOAC = direct oral anticoagulant; GAMIS = German-Austrian Aspirin Trial; GEMINI-ACS = Generic Evaluation of Management Intervention New Insights in Acute Coronary Syndromes; GLOBAL-LEADERS = A Clinical Study Comparing Two Forms of Antiplatelet Therapy After Stent Implantation; MI = myocardial infarction; OAC = oral anticoagulant; PAD = peripheral artery disease; PARIS = Persantine-Aspirin Reinfarction Study; PCI = percutaneous coronary intervention; PEGASUS-TIMI-54 = Prevention with Ticagrelor of Secondary Thrombotic Events in High-Risk Patients with Prior Acute Coronary Syndrome – Thrombolysis In Myocardial Infarction Study Group; STOP-DAPT2 = Short and Optimal Duration of Dual Antiplatelet Therapy-2; TARGET-FIRST = Evaluation of a Modified Anti-Platelet Therapy Associated With Low-dose DES Firehawk in Acute MI Patients Treated With Complete Revascularization Strategy; TICAB = Ticagrelor in Coronary Artery Bypass Grafting; TWILIGHT = Ticagrelor with Aspirin or Alone in High-Risk Patients after Coronary Intervention; WARIS = warfarin, aspirin, or both after MI.

risk of ischemic events.⁷¹ Deescalation can also be pursued by dose reduction. However, low-dose prasugrel monotherapy (3.75 mg daily) in Japanese patients very early after PCI (after a 20-mg loading dose) with ACS or at high bleeding risk seemed to increase 30-day MACE compared with DAPT.⁷⁷ Cumulative ischemic risk was low in most of these trials and may not be generalizable to higher-risk patients.

Antiplatelet therapy with PCI for CCD

Clopidogrel is the default P2Y₁₂ inhibitor after PCI for CCS. Although dedicated studies are lacking in this group of patients, ticagrelor or prasugrel can be considered in patients at high ischemic and low bleeding risk. Studies of shorter versus longer DAPT include a mix of patients with CCS and ACS, ranging from 40% to 70% with CCD. Shorter DAPT durations of 3 to 6 months followed by aspirin monotherapy versus 12 to 24 months of DAPT show no difference in ischemic events but lower risks of bleeding.⁷⁸⁻⁸² Trials of shorter-term DAPT followed by ticagrelor or prasugrel monotherapy after PCI in ACS also included patients being treated for CCD^{62,64,83-85} and showed lower ischemic and bleeding risk compared with longer DAPT with these potent agents.

CABG surgery

Two trials have assessed the use of DAPT after CABG surgery. The TACSI (Ticagrelor and Aspirin or Aspirin Alone after Coronary Surgery for Acute Coronary Syndrome) study was a pragmatic trial comparing ticagrelor plus low-dose aspirin versus aspirin alone in patients who had CABG surgery within 6 weeks of an ACS.⁸⁶ DAPT did not reduce the risk of death, MI, stroke, or repeat revascularization compared with aspirin alone at 1 year. Meta-analysis of observational and randomized trial data among patients having CABG surgery for ACS and non-ACS indications found lower cardiovascular death and vein graft failure but no difference in all-cause death, MI, or stroke, and higher rates of major bleeding with DAPT versus low-dose aspirin. A higher risk of bleeding with routine use also may be possible.^{87,88} Thus, DAPT may reduce the risk of vein graft failure after CABG surgery but has little effect on clinical outcomes except for increasing the risk of bleeding.

Consensus recommendations: antiplatelet therapy postrevascularization

- In patients with ACS undergoing PCI, prasugrel or ticagrelor are preferred as the P2Y₁₂ inhibitor of choice, with some data favoring prasugrel. Clopidogrel may be preferentially considered in patients >75 years of age due to the higher risk of major bleeding with prasugrel or ticagrelor.

- The default DAPT duration post-PCI is 6 months for CCD and 12 months for ACS. Aspirin should be maintained in the peri-PCI period. Shortened DAPT (1-3 mo) followed by P2Y₁₂ inhibitor monotherapy can reduce bleeding without increasing ischemic events. This approach is mainly applicable to ticagrelor (more robust data) or prasugrel, not clopidogrel or aspirin unless the patient is HBR. Caution is also advised in patients with higher ischemic risk, such as those presenting with STEMI.
- Guided or unguided deescalation from either ticagrelor or prasugrel to clopidogrel 1 month after ACS treated with PCI may be an option in selected patients to lower bleeding risk.
- DAPT reduces the risk of vein graft failure after CABG surgery but has little effect on clinical outcomes except for increasing the risk of bleeding.

LONG-TERM ANTITHROMBOTIC THERAPY BEYOND 1 YEAR AFTER ACUTE CORONARY EVENTS OR REVASCULARIZATION

The choice of antithrombotic therapy beyond 1 year after an ACS or coronary revascularization requires careful balancing of residual ischemic risk against bleeding risk. Although DAPT is standard during the acute period after ACS or PCI, evidence supporting long-term strategies beyond this period is more nuanced and increasingly individualized depending on risk profiles.

Aspirin monotherapy: placebo-controlled evidence and dosing

Aspirin has served as the foundation of secondary prevention for decades. Early randomized placebo-controlled trials conducted in post-MI populations—including Cardiff I and II, PARIS (Persantine-Aspirin Reinfarction Study), CDPA (Coronary Drug Project Aspirin), AMIS (Aspirin Myocardial Infarction Study), and GAMIS (German-Austrian Aspirin Trial)—demonstrated consistent reductions in recurrent MI and vascular mortality with long-term aspirin therapy compared with placebo despite heterogeneity in dose and study design.⁸⁹⁻⁹⁴ These findings were reinforced by the ATT (Antithrombotic Trialists') Collaboration meta-analyses, which pooled individual patient data from randomized trials and demonstrated that long-term antiplatelet therapy reduced the risk of serious vascular events by approximately 20% to 25% in secondary prevention populations,^{95,96} establishing aspirin as standard long-term antithrombotic therapy.

Subsequent trials evaluated the optimal aspirin dose. The largest and most pragmatic ADAPTABLE (Aspirin Dosing: A Patient-Centric Trial Assessing Benefits and Long-Term Effectiveness) trial, compared low-dose

aspirin (81 mg) versus standard aspirin (325 mg) in patients with established ASCVD and found no significant difference in death, MI, or stroke, or in major bleeding.⁹⁷ High rates of dose switching from 325 to 81 mg suggested superior tolerability of low-dose aspirin, supporting its preferential use for long-term therapy.⁹⁷

Aspirin versus prolonged DAPT

Extended DAPT beyond 1 year has been evaluated as a means of reducing late ischemic events, particularly after PCI. The DAPT (Dual Antiplatelet Therapy) trial showed that continuation of DAPT for 30 months after DES implantation reduced ST and spontaneous MI compared with aspirin alone, but significantly increased moderate or severe bleeding and was associated with increased noncardiovascular death.⁹⁸ In patients with previous MI not limited to stented segments, the PEGASUS-TIMI 54 (Prevention with Ticagrelor of Secondary Thrombotic Events in High-Risk Patients with Prior Acute Coronary Syndrome-Thrombolysis In Myocardial Infarction Study Group) trial evaluated prolonged ticagrelor therapy added to aspirin beyond 1 year after MI. Ticagrelor significantly reduced the composite of cardiovascular death, MI, or stroke compared with placebo, at the expense of increased major bleeding.^{99,112} These data support prolonged DAPT only in carefully selected patients with high ischemic risk and acceptable bleeding risk.

Aspirin versus P2Y₁₂ inhibitor monotherapy

Substitution of aspirin with P2Y₁₂ inhibitor monotherapy represents an alternative long-term strategy. Earlier trials such as CAPRIE (Clopidogrel versus Aspirin in Patients at Risk of Ischemic Events)¹¹³ demonstrated that clopidogrel was at least as effective as aspirin for secondary prevention, supporting its use as an alternative in aspirin-intolerant patients. Emerging data suggest that clopidogrel monotherapy may offer greater ischemic protection with similar or lower bleeding risk (potentially lower gastrointestinal and intracranial bleeding) than aspirin for secondary CAD prevention.^{24,114,115} In SMART-CHOICE 3 (Smart Angioplasty Research Team: Choice of Optimal Anti-Thrombotic Strategy in Patients Undergoing Implantation of Coronary Drug-Eluting Stents), clopidogrel reduced MACE (primarily MI) without increasing bleeding compared with low-dose aspirin among patients event-free after 6 to 12 months of DAPT post-PCI.¹⁰² Ten-year follow-up of the HOST-EXAM (Harmonizing Optimal Strategy for Treatment of Coronary Artery Stenosis-Extended Antiplatelet Monotherapy) trial showed a significant reduction in ischemic and bleeding events with clopidogrel monotherapy compared with aspirin monotherapy.¹⁰³ This topic deserves further research and longer-term follow-up, including whether lifelong

antiplatelet therapy for secondary prevention is necessary in all patients with CAD.¹¹⁶

Aspirin versus oral anticoagulation alone in patients without an indication for oral anticoagulation

The WARIS (Warfarin, Aspirin, or Both after MI)¹⁰⁶ and WARIS II trials demonstrated that warfarin monotherapy was at least as effective as aspirin in reducing recurrent ischemic events after MI, but with higher bleeding risk and management complexity.¹⁰⁷ These findings limited routine adoption of oral anticoagulation alone in stable coronary disease. In the modern era, the COMPASS (Rivaroxaban with or without Aspirin in Stable Coronary Disease) trial demonstrated that adding rivaroxaban 2.5 mg twice daily to low-dose aspirin significantly reduced MACE and total mortality rate in patients with stable CAD or PAD, despite a higher risk of major nonfatal bleeding. Important exclusion criteria were patients with previous stroke or previous major bleeding. COMPASS also demonstrated that rivaroxaban 5 mg twice daily alone did not significantly reduce MACE versus aspirin alone and increased major bleeding, arguing against full-dose DOAC as a substitute for aspirin beyond 1 year.¹¹⁷

Adding DOAC to DAPT has generally resulted in excess bleeding. The APPRAISE-2 (Apixaban for Prevention of Acute Ischemic Events 2) trial demonstrated increased major bleeding without ischemic benefit when apixaban was added to antiplatelet therapy after ACS.¹⁰⁹ The ATLAS ACS 2-TIMI 51 (Anti-Xa Therapy to Lower Cardiovascular Events in Addition to Standard Therapy in Subjects with Acute Coronary Syndrome-Thrombolysis In Myocardial Infarction 51) trial showed reduced ischemic events with low-dose rivaroxaban added to DAPT but with approximately 3-fold higher major bleeding.¹¹⁰ The GEMINI-ACS-1 (Generic Evaluation of Management Intervention New Insights in Acute Coronary Syndromes) trial explored replacing aspirin with low-dose rivaroxaban on a background of P2Y₁₂ inhibition and showed similar bleeding rates, although the study was not powered for ischemic efficacy.¹¹¹

Consensus recommendations: long-term antiplatelet therapy after ACS or MI

- Beyond 1 year after ACS, emerging data suggest that clopidogrel monotherapy may offer greater ischemic protection with similar or lower bleeding risk compared with aspirin monotherapy for secondary CAD prevention.^{24,114,115}
- Intensified strategies, including prolonged DAPT, P2Y₁₂ inhibitor monotherapy, or anticoagulant-based regimens, may be appropriate in selected high-risk patients but ischemic benefit should be consistently weighed against risk of increased bleeding. Individualized assessment of ischemic and bleeding risk

remains central to long-term antithrombotic decision-making.

- Rivaroxaban 2.5 mg twice daily along with aspirin 81 mg daily can be considered for long-term use in select high-risk patients with stable CAD or PAD who do not have a history of previous major bleeding or cerebrovascular accident.

ANTIPLATELET THERAPY IN SPECIAL POPULATIONS

PAD

Acute management

After an endovascular intervention for PAD, antiplatelet therapy is often intensified to prevent thrombotic complications. Poststenting, DAPT with aspirin plus a P2Y₁₂ inhibitor is typically considered for 1 to 6 months and is a ACC/AHA Class 2a recommendation in the 2024 ACC/AHA PAD guidelines.²⁷ Shorter-course DAPT (≤ 1 month) followed by antiplatelet monotherapy may be sufficient in patients undergoing surgical bypass, especially with a prosthetic graft. The VOYAGER-PAD (Vascular Outcomes Study of ASA Along with Rivaroxaban in Endovascular or Surgical Limb Revascularization for PAD) trial¹¹⁸ evaluated low-dose rivaroxaban (2.5 mg twice daily) plus aspirin in patients within 10 days of lower-extremity revascularization who were not on other anticoagulants or planned for P2Y₁₂ inhibitor use. Patients with HBR were excluded. The regimen reduced the composite of MACE and major acute limb events versus aspirin alone, primarily due to fewer major acute limb events. This is thus an effective strategy as well. If a patient with PAD requires full-dose anticoagulation for another indication, such as atrial fibrillation (AF), adding single antiplatelet therapy on top of anticoagulation after revascularization is considered a reasonable approach. Prolonged triple therapy (anticoagulation + DAPT) is not recommended due to HBR. Both aspirin and clopidogrel are reasonable choices for antiplatelet therapy in this patient population, although the CAPRIE trial suggests that clopidogrel may be a better choice.¹⁰¹ Ticagrelor showed comparable safety and efficacy to clopidogrel in the EUCLID (Examining Use of Ticagrelor in PAD) trial, although rates of adverse events were higher with ticagrelor.¹¹⁹

Chronic management

Patients with symptomatic PAD should be on long-term single antiplatelet therapy to reduce the risk of MACE. Acceptable options are aspirin 75 to 325 mg daily or clopidogrel 75 mg daily (both ACC/AHA Class 1 and ESC Class I recommendations), although the CAPRIE trial suggests that clopidogrel may be a better choice in this patient

population.²⁷ Routine long-term DAPT (aspirin + clopidogrel) in stable PAD is not advised when no recent intervention has been performed because the benefits are uncertain and the bleeding risk is increased. In asymptomatic PAD (eg, low ankle-brachial index without symptoms), antiplatelet therapy may be considered to reduce MACE, but the evidence base is less robust. In contrast to antiplatelet therapy, anticoagulation alone is not indicated in patients with PAD, because it is ineffective and potentially harmful. Based on the results of COMPASS, including the PAD subset,¹²⁰ and VOYAGER-PAD, adding rivaroxaban 2.5 mg twice daily to aspirin is a reasonable long-term strategy in patients with PAD to reduce MACE and major adverse limb events, particularly among high-risk patients (eg, those with previous amputation or revascularization, polyvascular disease, diabetes, or renal insufficiency) who have acceptable bleeding risk.

Cerebrovascular disease: stroke or TIA

Acute management

Acute ischemic strokes can occur in the setting of underlying atherosclerosis or result from embolism typically due to thrombus arising in the left ventricle or left atrium or as a result of paradoxical embolism. The 2026 AHA/American Stroke Association guideline for management of acute ischemic stroke recommends starting aspirin within 48 hours of stroke onset for acute ischemic stroke that is not cardioembolic, after brain imaging to rule out hemorrhage.²⁶ If thrombolytic therapy was administered, aspirin initiation is often delayed for 24 hours, although the risk remains unclear.¹²¹ A clear bleeding risk is apparent with use of aspirin in this patient population (concurrently or within 90 minutes) after the start of thrombolytic therapy.¹²¹ In patients with minor ischemic stroke (National Institutes of Health Stroke Scale [NIHSS] score of ≤ 3) or high-risk TIA (ABCD² score of ≥ 4), evidence supports initiating DAPT, typically with aspirin plus clopidogrel (300-600 mg loading dose followed by 75 mg daily) over aspirin monotherapy within 12 to 24 hours of onset and continuing for approximately 21 days to reduce the risk of early recurrent stroke. After 21 days, DAPT is typically deescalated to antiplatelet monotherapy to minimize bleeding risk while maintaining protection against 90-day recurrent stroke.¹²²⁻¹²⁷ Ticagrelor for 30 days can be considered as part of the DAPT regimen instead of clopidogrel, although this strategy results in a higher bleeding risk compared with aspirin monotherapy.¹²⁸ Ticagrelor monotherapy is not superior to aspirin among patients with minor ischemic stroke (NIHSS score of ≤ 3) or high-risk TIA (ABCD² score ≥ 4) and is therefore not indicated in this patient population unless they have aspirin

intolerance.¹²⁹ The use of DAPT among patients who receive thrombolysis or have more severe strokes (NIHSS score of >5) is unclear.

For patients with stroke due to AF or other cardioembolic sources, anticoagulation rather than antiplatelet therapy is recommended after the acute period.¹³⁰ The optimal timing of anticoagulation in this patient population is unclear. In the OPTIMAS (Optimal Timing of Anticoagulation After AIS with Atrial Fibrillation) trial, early DOAC therapy (≤ 4 days of onset) was noninferior to delayed DOAC therapy (7-14 days of onset)¹³¹ and can therefore be considered among patients at low bleeding risk. Patients with extracranial carotid or vertebral dissections can be treated with either anticoagulation or antiplatelet therapy for at least 3 months.¹³²

Chronic management

In the chronic phase, patients with atherosclerotic or cryptogenic ischemic stroke should be on an antiplatelet agent rather than anticoagulation for secondary prevention.²⁶ Acceptable regimens include aspirin 81 mg daily, clopidogrel 75 mg daily, or aspirin combined with extended-release dipyridamole (eg, Aggrenox).¹²³ Long-term DAPT is not recommended.¹³³ Prolonged aspirin with clopidogrel is contraindicated, especially after lacunar strokes, based on the results of the SPS3 (Secondary Prevention of Small Cortical Strokes) trial^{134,135} where DAPT with clopidogrel and aspirin did not reduce recurrent strokes but significantly increased bleeding and death.

Detection of AF, even paroxysmal, should prompt anticoagulation, and any long-term antiplatelet can often be discontinued unless another indication exists. Conversely, if a patient who experiences a stroke requires anticoagulation but has a separate indication for antiplatelet therapy (eg, recent coronary stenting or carotid stenting), combination therapy might be needed short term. Among patients with cryptogenic ischemic strokes that are potentially embolic (25%-30% of all ischemic strokes), screening for AF with implantable monitors is more sensitive compared with clinical monitoring¹³⁶ and may have therapeutic implications.^{137,138}

AF or venous thromboembolism with concurrent antiplatelet needs

Patients with AF or venous thromboembolism often require long-term anticoagulation to prevent cardioembolic stroke or recurrent venous thromboembolism. When such patients also have an indication for antiplatelet therapy, careful balancing of thrombotic versus bleeding risk is required. An emphasis is on a “less is more” approach to combination therapy.²⁷

In the acute phase after coronary stent implantation, triple therapy with an OAC plus both aspirin and a P2Y₁₂ inhibitor should be used for the shortest duration possible and is generally not advised for routine long-term use due to HBR.^{17,18,25} When necessary, aspirin is typically prescribed for 1 to 4 weeks. When triple therapy is used, a low dose of enteric-coated aspirin (<100 mg) and clopidogrel are preferred to minimize bleeding risk,²⁵ because prasugrel and ticagrelor confer a higher bleed risk. After the first 1 to 4 weeks of triple therapy, a strategy of double antithrombotic therapy with an OAC and a P2Y₁₂ inhibitor is advised. Typically, this means a DOAC (preferred over warfarin in eligible patients) combined with clopidogrel because it has been most widely studied. The duration of double antithrombotic therapy will depend on the indication for anticoagulation and bleeding risk. After a standard course of double antithrombotic therapy, anticoagulation monotherapy is typically preferred because it has been shown to reduce bleeding without an increase in ischemic events.¹³⁹⁻¹⁴²

If a patient on chronic aspirin (eg, for stable CAD beyond 1 year from PCI or ACS) develops new-onset AF or acute deep vein thrombosis or pulmonary embolism, antiplatelet therapy can similarly be discontinued and the patient switched to appropriate dose anticoagulation. Anticoagulant monotherapy will address stroke risk from AF and likely affords some protection against coronary events. In patients with provoked venous thromboembolism requiring a finite course of anticoagulation therapy, antiplatelet therapy should be restarted once the anticoagulant is stopped.

Valvular heart disease

Among patients with valvular heart disease, especially those with prosthetic heart valves, antiplatelet therapy may overlap with anticoagulation. These patients often require indefinite anticoagulation with a vitamin K antagonist (warfarin). Goals for target international normalized ratio values depend on valve type and position (eg, for mechanical aortic valve usually target international normalized ratio 2.5; mechanical mitral valve international normalized ratio 3.0).¹⁴² Antiplatelet therapy added to warfarin in patients with mechanical valves for atherosclerotic disease or risk has a weak recommendation (ACC/AHA Class 2b) in the 2020 ACC/AHA guideline for the management of patients with valvular heart disease.¹⁴³ A special case is the On-X mechanical aortic valve (Artivion, Kennesaw, Georgia): among patients with mechanical On-X aortic valve replacement and no thromboembolic risk factors, a lower international normalized ratio range (1.5-2.0) along with aspirin 75 to 100 mg daily can be reasonably pursued ≥ 3 months after surgery. DOACs are contraindicated for use in patients

with mechanical valves.¹⁴³ Bioprosthetic valves do not require long-term warfarin, but antithrombotic therapy is still recommended. In patients with a bioprosthetic aortic or mitral valve, whether replaced via surgical (surgical aortic valve replacement or mitral valve replacement) or transcatheter (transcatheter aortic valve implantation) approach, daily aspirin 75 to 100 mg is recommended in the absence of other indications for anticoagulation. Recent evidence has shown no benefit to routine DAPT or OAC after transcatheter aortic valve implantation.¹⁴⁴⁻¹⁴⁷ For surgical bioprosthetic valves, the 2020 ACC/AHA valve guidelines recommend warfarin (international normalized ratio ~2.5) as a reasonable option for the first 3 to 6 months after surgical bioprosthetic valve replacement to reduce risk of valve thrombosis (ACC/AHA Class 2a). This is often done particularly for bioprosthetic mitral valves, which have higher thrombotic risk. Warfarin should then be discontinued in favor of aspirin after the initial postoperative period. If the patient has another indication for anticoagulation (eg, the patient with a bioprosthetic valve is in AF), that anticoagulation should be continued; in such cases, the routine aspirin might be omitted or used only short term, depending on bleeding risk. In a patient with a mechanical valve who also has coronary disease, the need for warfarin is clear, and adding aspirin is typically advised if tolerated.

Consensus recommendations: PAD, stroke, TIA, valve disease, and patients requiring anticoagulation

- Antiplatelet therapy has an important role in the acute and chronic management of PAD and cerebrovascular disease, similar to CAD. However, compared with CAD, shorter durations of DAPT (1-6 mo) are typically recommended postrevascularization.
- Adding rivaroxaban 2.5 mg twice daily to low-dose aspirin is a reasonable long-term strategy in patients with PAD to reduce MACE and major adverse limb events, particularly among high-risk patients who have acceptable bleeding risk.
- In patients with minor ischemic stroke (NIHSS score of ≤ 3) or high-risk TIA (ABCD² score of ≥ 4), early initiation of DAPT with aspirin plus clopidogrel within 12 to 24 hours, continued for 21 days, reduces early recurrent stroke compared with aspirin alone, after which therapy is deescalated to antiplatelet monotherapy to limit bleeding risk. The use of DAPT among patients who receive thrombolysis or have more severe strokes (NIHSS score of > 5) is unclear.
- Triple therapy among patients needing OAC and antiplatelet therapy should be avoided. Aspirin can typically be discontinued within 1 month of PCI; clopidogrel and DOACs are the favored agents for antithrombotic therapy in these patients. Long-term

management (beyond 1 year) can include DOACs as monotherapy.

- Patients receiving bioprosthetic aortic or mitral valves can be managed with single antiplatelet therapy (typically low-dose aspirin) long term.

OTHER CONSIDERATIONS FOR MANAGEMENT OF ANTIPLATELET THERAPY

Discontinuing DAPT before elective noncardiac surgery

Surgery is associated with a prothrombotic state, and patients face the highest risk of ST in the first 4 to 6 weeks after PCI. The balance of risk of hemorrhage from noncardiac surgery and ST, MI, and death determine strategies for managing antiplatelet therapy in the perioperative period. The 2024 ACC/AHA guideline for perioperative cardiovascular management for noncardiac surgery presents a recommended strategy of deferring elective surgery for ≥ 6 months after PCI for CCS and 12 months after PCI for ACS.² After this time, P2Y₁₂ inhibitors can be stopped, but low-dose aspirin should be continued. Due to the risk of serious complications associated with bleeding, some types of surgery (eg, brain or spinal surgery) may require cessation of aspirin 7 days preoperatively. Stopping antiplatelet therapy poses an elevated risk of MI or ST, and these risks must be carefully balanced against the risks of perioperative hemorrhage. The recommended timing of stopping P2Y₁₂ inhibitors before surgery is based on the half-life of these drugs. Clopidogrel and ticagrelor are stopped 5 days before surgery and prasugrel 7 days before surgery.¹⁴⁸

Strategies for urgent noncardiac surgery

Urgent or emergency noncardiac surgery may be required in patients who have undergone PCI for CCS within the past 6 months or PCI for ACS within the past 12 months. Although some studies suggest that shorter DAPT durations (eg, 3 mo) may yield outcomes similar to longer regimens, these strategies have not been tested in the prothrombotic setting of noncardiac surgery, and therefore, the benefits of early surgery should be balanced by the potential risk of ST. If timely surgery is necessary within 3 months of stent implantation, bridging with intravenous antiplatelet agents should be considered. Conversely, when surgery is needed > 3 months after PCI, discontinuation of the P2Y₁₂ agent (with continued aspirin) with or without bridging is reasonable. Factors to be considered to determine the need to bridge include the time from PCI to cessation of the P2Y₁₂ agent, stent location, or stent size and number. When possible, surgery should be delayed for 5 to 7 days to allow P2Y₁₂ inhibitor washout. If surgery cannot be delayed for

TABLE 5 Perioperative Management of Antiplatelet Agents^{150,151}

Action	Timing
Hold prasugrel	7 d before
Hold clopidogrel or ticagrelor	5 d before
Cangrelor IV bridge after stopping oral P2Y ₁₂ inhibitor	From ~5 d before until surgery
Eptifibatide or tirofiban Short-acting IV antiplatelet alternative for bridging	From ~5 d before until hours before surgery
SURGERY	
Restart oral P2Y ₁₂ inhibitor Restart once hemostasis is achieved	4–6 h after surgery

IV = intravenous.

adequate washout and bleeding risk is high, reversal of the P2Y₁₂ inhibitor effect may be considered.

Antiplatelet bridging

Bridging refers to stopping oral P2Y₁₂ inhibitors and treating with an intravenous antiplatelet drug before and sometimes after surgery until the oral P2Y₁₂ inhibitor is restarted. Bridging is generally used when surgery is needed within 3 months of a PCI, and concerns are present regarding heightened risk of stent thrombosis during the time the P2Y₁₂ inhibitor is being held. Only limited data on using intravenous antiplatelet agents in this manner are available. The 3 agents that have been described for this purpose are the GPIs (eptifibatide or tirofiban) or cangrelor.¹⁴⁸ Cangrelor has a shorter onset and offset time and can be stopped a shorter time period before surgery; dosing is based on the BRIDGE (Bridging Antiplatelet Therapy With Cangrelor in Patients Undergoing Cardiac Surgery) trial (0.75 mcg/kg/min infusion without a bolus, stopped 1 h before surgery).¹⁴⁹ However, the cost of these agents may differ between institutions and may determine which is the most cost-effective to use. A suggested plan for antiplatelet bridging is shown in [Table 5](#).¹⁵⁰

Reversal of P2Y₁₂ inhibitors

Limited data are available on P2Y₁₂ inhibitor reversal for emergency surgery with high hemorrhagic risk. No direct reversal agents are available for clopidogrel or prasugrel. Bentracimab is a monoclonal antibody fragment that is specific for ticagrelor and has been studied in the REVERSE-IT (Bentracimab in Ticagrelor-treated Patients With Uncontrolled Bleeding or Requiring Urgent Surgery or Invasive Procedure) trial,¹⁵¹ but it is not yet available for clinical use. Platelet transfusions may reverse some of the effects of clopidogrel or prasugrel but generally have little effect on ticagrelor. Prothrombotic agents have been suggested (eg, tranexamic acid or desmopressin acetate [DDAVP]), but these are primarily based on

ex vivo or animal testing, and their effect on human beings in this setting has not been evaluated. Recommendations from international groups¹⁵² suggest that platelet transfusion may influence platelet function if given ≥6 hours after the last oral dose of clopidogrel or prasugrel, and 24 hours after the last oral dose of ticagrelor, with 2 to 4 units of platelets suggested. Higher doses of platelet transfusion may be required for prasugrel and ticagrelor than clopidogrel.

Consensus recommendations: perioperative management of antiplatelet therapy

- The 2024 AHA/ACC guideline for perioperative cardiovascular management for noncardiac surgery recommends strategies of deferring elective surgery ≥6 months after PCI for CCS and 12 months after PCI for ACS.² After this time, P2Y₁₂ inhibitors can be stopped but low-dose aspirin should be continued. In the current era, shorter DAPT durations can be considered in carefully selected patients.
- The recommended timing of stopping P2Y₁₂ inhibitors before surgery is based on the half-life of these drugs. Clopidogrel and ticagrelor are stopped 5 days before surgery and prasugrel 7 days before surgery.
- Urgent noncardiac surgery may require cessation of P2Y₁₂ agents and bridging with a shorter-acting intravenous antiplatelet agent, or the use of platelet transfusion for more urgent cases (eg, intracranial bleeding).

Monitoring of patients on antiplatelet therapies

Although clinical trials have largely used a “one-size-fits-all” strategy for antiplatelet therapy, individual pharmacodynamic responses, particularly to aspirin and clopidogrel, vary substantially. Certain populations, including those with diabetes or renal dysfunction, may be at increased risk of antiplatelet non- or hypo-responsiveness.¹⁰ Further, the generation of clopidogrel’s active metabolite is variable and unpredictable, resulting in significant interpatient differences in on-treatment platelet reactivity. Both high and low platelet reactivity have been observed, each associated with adverse outcomes.¹⁵⁴⁻¹⁵⁵

The historical gold standard for assessing platelet reactivity is light transmission aggregometry, which measures platelet aggregation for aspirin and P2Y₁₂ inhibitors. Aspirin resistance was defined as ≥20% on-treatment aggregation with arachidonic acid,^{156,157} while clopidogrel resistance was defined as ≤10% change in ADP-induced aggregation from baseline.^{158,159} Because of its tedious nature, poor reproducibility and interindividual variability,¹⁶⁰ light transmission aggregometry has largely been replaced by other in vitro modalities such as PFA-100 (aspirin), VerifyNow (Werfen LLC, Bedford, MA) (aspirin and P2Y₁₂ inhibitors), and vasodilator-stimulated

phosphoprotein phosphorylation (assesses P2Y₁₂ receptor inhibition). In addition, an aspirin reaction unit ≥ 550 on VerifyNow is generally considered evidence of aspirin resistance.^{153,161-165}

In contrast to laboratory-based antiplatelet resistance, clinical resistance refers to the occurrence of thrombotic events despite antiplatelet therapy, reflecting ineffective or incomplete platelet inhibition.¹⁶³ However, the most common cause is medication noncompliance,¹⁶⁶⁻¹⁶⁹ and some thrombotic events may arise through non-platelet-mediated pathways not affected by antiplatelet therapy (eg, a mechanical issue such as stent underexpansion).

Genetic polymorphisms can contribute at least in part, to the variability in clopidogrel response.¹⁷⁰⁻¹⁷² The CYP2C19 enzyme mediates both metabolic steps required for conversion of clopidogrel into its active metabolite, but the gene encoding this enzyme is highly polymorphic, with variants associated with reduced (loss-of-function) or increased (gain-of-function) activity.¹⁷¹⁻¹⁷³ In contrast, prasugrel and ticagrelor have minimal or no dependence on CYP2C19 for their antiplatelet effects.¹⁷⁴ CYP2C19*1 denotes normal (“wild-type”) function (prevalence ~25%-60%, depending on ancestry),¹⁷⁵ whereas *2 and *3 are the most common loss-of-function alleles, and *17 is a gain-of-function allele.¹⁷⁶ Carriers of loss-of-function alleles have reduced formation of clopidogrel’s active metabolite, higher rates of high on-treatment platelet reactivity, and increased thrombotic risk after PCI.^{170,172,174,177,178} Approximately 2% to 15% of individuals are poor metabolizers,¹⁷⁷ defined by 2 loss-of-function alleles (eg, *2/*2, *3/*3, *2/*3). Because the clinical impact of gain-of-function alleles remains less certain, CYP2C19*2 and *3 are typically included in standard pharmacogenomic testing panels.

Collectively, these observations have led to the premise of a “precision medicine” approach for patients undergoing PCI, whereby treatment regimens, including antiplatelet therapy, could be individualized to optimize safety and efficacy, including escalation and deescalation of DAPT. The POPULAR Genetics Trial (Cost-effectiveness of Genotype Guided Treatment With Antiplatelet Drugs in STEMI Patients: Optimization of Treatment) showed that a peri-PCI CYP2C19 genotype-guided strategy for selecting oral P2Y₁₂ inhibitor therapy was noninferior for ischemic outcomes and superior for bleeding outcomes compared with standard treatment with ticagrelor or prasugrel in patients undergoing primary PCI.⁷³ However, data to support this approach remain mixed.¹⁷⁹⁻¹⁸¹ The 2025 ACC/AHA ACS guidelines note that deescalation of DAPT intensity from prasugrel or ticagrelor to clopidogrel may be guided by platelet function or genotyping assays, while also acknowledging that unguided deescalation strategies may be similarly effective.^{17,21}

Drug-drug interactions

Aspirin

Ibuprofen and other nonsteroidal anti-inflammatory drugs, such as indomethacin and naproxen, are competitive inhibitors of the catalytic site of COX-1,^{182,183} leading to reversible inhibition of thromboxane A₂ formation and preventing aspirin to access the serine residue of COX-1. Concomitant administration of aspirin and ibuprofen, especially when ibuprofen is given before aspirin, significantly reduces the antiplatelet effect of aspirin as measured by serum thromboxane B₂ levels.¹⁸⁴ Long-term concomitant use increases the risk of gastrointestinal bleeding and renal dysfunction¹⁸⁵ and results in an increased risk of all-cause death and cardiovascular death.¹⁸⁶ This effect was not observed for acetaminophen or COX-2 selective inhibitors.¹⁸⁴

P2Y₁₂ inhibitors

Several drugs that are CYP3A4 inhibitors (eg, ketoconazole, itraconazole, nifedipine, amlodipine)¹⁸⁷⁻¹⁸⁹ or are metabolized by CYP3A4 (eg, phenprocoumon)¹⁹⁰ have been shown to decrease the antiplatelet effect of clopidogrel.¹⁹⁰⁻¹⁹² Conversely, substances known to induce CYP activity (eg, rifampin)¹⁹³ increase platelet inhibition by clopidogrel. The clinical relevance of these interactions may be marginal. Nirmatrelvir-ritonavir does not interact with aspirin but has important interactions with P2Y₁₂ inhibitors. It inhibits CYP3A4 and weakly induces CYP2C19. Although it reduces bioavailability of both clopidogrel and prasugrel, the clinically meaningful effect is mainly with clopidogrel. In patients with recent PCI or high thrombotic risk, it is generally preferable to avoid nirmatrelvir-ritonavir or switch from clopidogrel to prasugrel during treatment. It also increases bioavailability of ticagrelor, which may increase bleeding risk; switching to prasugrel may similarly be considered.¹⁹⁴ Another important consideration is proton pump inhibitors (PPIs), which are known CYP2C19 substrates and show different potency to inhibit CYP2C19. For example, omeprazole shows a greater affinity for CYP2C19 and therefore has a greater potential for drug-drug interactions mediated by CYP2C19.¹⁹² Concomitant use of clopidogrel with omeprazole but not pantoprazole has consistently been shown to attenuate the antiplatelet effect of clopidogrel in ex vivo studies.^{189,195,196} Nonetheless, a double-blind, placebo-controlled randomized trial demonstrated no significant difference in ischemic events between omeprazole and placebo in patients treated with clopidogrel, while PPI use significantly reduced gastrointestinal bleeding risk.¹⁹⁷ Moreover, post hoc analyses from multiple randomized trials have shown no increase in ischemic events when a clinically indicated PPI is used concomitantly with clopidogrel.^{198,199} Importantly, the antiplatelet effects and clinical efficacy

of ticagrelor and prasugrel are not meaningfully affected by PPI coadministration.²⁰⁰ Therefore, the 2025 ACC/AHA ACS guideline recommends PPI therapy for patients with ACS at increased bleeding risk who are receiving DAPT or oral anticoagulation.^{17,25}

Consensus recommendations: monitoring of patients on antiplatelet therapy

- Considerable variability exists in patients' response to antiplatelet therapy, particularly with clopidogrel. Although routine platelet function testing and genotyping are not universally recommended, they may have a role in guiding deescalation of P2Y₁₂ inhibitor therapy.
- Routine PPI therapy is recommended for patients at high risk of bleeding or for those requiring concomitant anticoagulant use. Concerns that concomitant omeprazole reduces clopidogrel efficacy have not been proved clinically.

Antiplatelet nonadherence

Medication adherence with antiplatelet therapy remains a major challenge: as early as 6 weeks after hospitalization for MI, nearly one-third of patients can be suboptimally adherent to prescribed cardiovascular therapies including antiplatelet therapy,²⁰¹⁻²⁰⁴ with rates increasing further by 1 year.²⁰⁴⁻²⁰⁷ Nonadherence to antiplatelet therapy is a key risk factor for ST, especially within the first few months, and is associated with increased all-cause death.²⁰⁴⁻²¹² In the PARIS-2 (Patterns of Non-Adherence to Anti-Platelet Regimen in Stented Patients) registry, disruption of antiplatelet therapy due to nonadherence or bleeding occurred in 2.1% at 30 days, 9.8% at 1 year, and 14.4% at 2 years, and was associated with a higher adjusted risk of MACE, particularly within the first 30 days. In contrast, continued DAPT use, or discontinuation based on physician recommendation or temporary interruption for surgery, was not associated with an increased risk of adverse events.²⁰⁴

A few important determinants of nonadherence deserve further discussion. Older patients are particularly vulnerable, due to issues such as difficulty hearing, comprehending, and remembering instructions; polypharmacy; tolerating adverse effects of medication; and drug-drug interactions.^{209,213-215} Another important risk factor is socioeconomic status, with lower socioeconomic status being associated with lower adherence despite a higher prevalence of cardiovascular risk factors.²¹⁶⁻²¹⁸ Higher prescription drug copayments and medication unaffordability are associated with lower adherence to antiplatelet therapy^{219,220} and an increased risk of rehospitalization among patients with ACS.²²⁰ Generic ticagrelor has reduced costs, but it can still be more

expensive than generic prasugrel or clopidogrel.²²¹ It is estimated that 1 of 4 low-income families with a member with ASCVD, including those with insurance coverage, experience a significant financial burden, and one-tenth experience a catastrophic financial burden due to out-of-pocket expenses that accumulate over time. Of these, nearly 40% of out-of-pocket expenditures were related to medication use.²²² Despite this, in the ARTEMIS (Affordability and Real-World Antiplatelet Treatment Effectiveness After Myocardial Infarction Study) trial, providing vouchers to lower copayments for clopidogrel or ticagrelor significantly improved patient adherence at 1 year but did not lead to a reduction in MACE compared with usual care.²²³ These findings suggest that copayment reduction or elimination may be most effective when implemented as part of a multipronged strategy to improve medication adherence and clinical outcomes, rather than as a standalone solution. This may be reflected in higher rates of nonadherence in Black versus White patients and patients with greater frailty.²⁰⁷

Based on patient interviews, occurrence of adverse effects interfering with daily life was the most common reason for treatment discontinuation.^{224,225} Bleeding is an important adverse effect and warrants closer monitoring, especially among those that are higher risk for bleeding, those on more potent and longer duration DAPT, or both. Deescalation should be considered when necessary. For ticagrelor, dose reduction to the 60 mg twice daily dose after the acute post-MI phase should be considered among patients needing longer-term DAPT, because the 60-mg dose has been found to have similar efficacy, but significantly lower nonadherence rates, mostly attributed to less bleeding and dyspnea compared with the 90-mg dose in the PEGASUS-TIMI 54 trial.¹⁰¹ Additionally, the use of ticagrelor in patients on oral anticoagulants has not been thoroughly investigated,²²⁶ and therefore, ticagrelor should be used with caution in combination antithrombotic regimens. Prasugrel should be avoided altogether in these patients.²²⁷ The concomitant use of nonsteroidal anti-inflammatory drugs continues to be a common practice and should be minimized. Although significant bleeding is an understandable cause, even nuisance bleeding such as epistaxis may prompt medication discontinuation.²²⁸ Among patients taking ticagrelor, ticagrelor-related dyspnea or conduction abnormalities are important reasons for premature discontinuation.^{215,229}

Some of these issues may be mitigated by early follow-up postdischarge (ideally within 6 weeks),²³⁰ with the aim of avoiding and managing minor adverse events or providing relevant alternative options to maintain therapy adherence. Culturally appropriate, collaborative, and clear communication are vitally important.²³¹⁻²³³ In a systematic review of 33 studies, almost all interventions

that were effective for long-term care were complex, including combinations of more convenient care, information, counseling, reminders, self-monitoring, reinforcement, family therapy, and other forms of additional supervision or attention. Even the most effective interventions had modest effects.²³⁴ A polypill strategy has shown promise for secondary prevention in patients with ASCVD, with some formulations combining aspirin, a statin, and antihypertensive agents. Small studies, mainly evaluating pharmacokinetics, have shown improved adherence with fixed-dose polypill combinations of aspirin and clopidogrel.^{235,236} In this context, ticagrelor may be considered less favorable than clopidogrel or prasugrel due to its twice-daily dosing requirement. Overall, the polypill approach represents a promising avenue for research that warrants further careful evaluation.

Consensus recommendations: adherence to antiplatelet therapy

- Medication adherence to DAPT and antiplatelet therapy remains a major challenge after MI, with non-adherence, particularly early post-PCI, associated with a higher risk of stent thrombosis and MACE.
- Factors contributing to poor adherence include older age, polypharmacy, socioeconomic barriers, high drug costs, and medication-related adverse effects such as bleeding or dyspnea.
- Strategies to improve adherence may include early postdischarge follow-up, culturally sensitive patient education, addressing social drivers of health such as financial burden, close monitoring for bleeding and other adverse effects, dose adjustment or deescalation of therapy when appropriate, and potentially the use of fixed-dose polypill combinations. Multifaceted and multilevel (ie, patient, family, healthcare system) approaches are essential.

EVIDENCE GAPS AND FUTURE DIRECTIONS

One of the biggest limitations of antiplatelet medications is the increased risk of bleeding. Various novel agents are being tested; an ideal antiplatelet drug would selectively inhibit thrombosis but leave hemostatic mechanisms

intact. As our understanding of platelet biology has advanced, drugs focused on preventing platelet adhesion and activation using novel therapeutic approaches are being actively pursued. These include drugs targeting cell surface receptors (eg, P2Y₁ and P2Y₁₂ inhibitor, GPVI, vWF-GPIb-GPIX-GPV, PAR4), or those focused on intracellular kinases and enzymes necessary for intracellular signaling pathways (eg, PI3K β , PDK1, BTK, annexin 7, PDI).²³⁷ Selatogrel is a promising novel P2Y₁₂ inhibitor that provides sustained and reversible P2Y₁₂ platelet inhibition with an acceptable safety profile in patients with CCS or ACS. It can be administered subcutaneously, and the SOS-AMI (Selatogrel Outcome Study in Suspected Acute Myocardial Infarction) trial is investigating whether self-injection of this agent among high-risk patients experiencing MI symptoms can improve cardiovascular outcomes.²³⁸ Similarly, zalunfiban is a subcutaneously administered GPI that showed promising results for prehospital utilization among patients with suspected STEMI in the CELEBRATE (Zalunfiban at First Medical Contact for ST-Elevation Myocardial Infarction) trial.²³⁸ Other areas of investigation include refining current algorithms for DAPT use after DES implantation, and defining the optimal duration of DAPT after drug-coated balloon therapy. The promise of tailored or precision medicine remains elusive, although artificial intelligence-based approaches may help enable this in the future.

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APPENDIX 1. AUTHOR RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (RELEVANT)— ANTIPLATELET THERAPY IN THE MANAGEMENT OF ATHEROSCLEROTIC CARDIOVASCULAR DISEASE: 2026 ACC SCIENTIFIC STATEMENT

Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Dharam J. Kumbhani, <i>Chair</i>	Section Chief, Director of Interventional Cardiology; Jim and Norma Smith Distinguished Chair for Interventional Cardiology, UT Southwestern Medical Center	None	None	None	None	None	None
C. Michael Gibson	Chief Executive Officer, Baim and PERFUSE Research Institutes; Professor of Medicine, Harvard Medical School	■ Anthos Therapeutics ■ Bayer* ■ BCRI ■ Boehringer Ingelheim ■ Cadrenal Therapeutics ■ Cardiovascular Research Foundation ■ CeleCor Therapeutics* ■ CSL Behring* ■ Daiichi Sankyo* ■ Faraday ■ HeartFlow ■ Janssen Pharmaceuticals ■ Johnson & Johnson* ■ Novo Nordisk ■ Roche ■ Somahlution/ Marizyme*	None	None	■ Bristol-Myers Squibb† ■ SCAD Alliance†	■ Amgen‡ ■ Baim Institute* ■ Beren‡ ■ Boehringer Ingelheim‡ ■ Boston Scientific‡ ■ CeleCor*‡ ■ Cleerly‡ ■ CSL Behring‡ ■ Daiichi Sankyo‡ ■ Duke Clinical Research Institute‡ ■ Janssen Pharmaceuticals* ■ Johnson & Johnson* ■ Merck & Co.‡ ■ Novo Nordisk*‡ ■ PhaseBio*‡ ■ Somahlution*‡	None
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Freek W.A. Verheugt	Professor of Cardiology, Heart-Lung Centre of the University Medical Centre of Nijmegen; Chairman of the Department of Cardiology, Onze Lieve Vrouwe Gasthuis	None	None	None	None	None	None

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*Significant relationship.

†No financial benefit.

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ACC = American College of Cardiology; UT = University of Texas.

APPENDIX 2. PEER REVIEWER RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (COMPREHENSIVE)—ANTIPLATELET THERAPY IN THE MANAGEMENT OF ATHEROSCLEROTIC CARDIOVASCULAR DISEASE: 2026 ACC SCIENTIFIC STATEMENT

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Rhian E. Davies	Systems Director, Interventional Cardiology; Director, Complex Coronary Program, WellSpan Health	■ Avinger ■ Boston Scientific ■ Medtronic ■ Rampart‡ ■ Shockwave ■ Teleflex ■ Terumo	None	None	None	None	None
Renato D. Lopes	Professor of Medicine; Member, Clinical Research Institute, Duke University	■ Bayer ■ Boehringer Ingelheim ■ Bristol Myers Squibb* ■ Novartis ■ Novo Nordisk ■ Pfizer*	None	None	■ Alnylam* ■ Basilea* ■ Bayer* ■ Bristol Myers Squibb* ■ Cytokinetics* ■ GlaxoSmithKline* ■ Ionis* ■ Novartis* ■ Pfizer* ■ Priovant* ■ Regeneron* ■ Roche* ■ Theravance Biopharma* ■ VeraDermics*	None	None

*Significant relationship. †Clinical trial enroller: Relationship with this company is limited to enrolling patients in clinical trials. This disclosure was entered under the Clinical Trial Enroller Category in the ACC's disclosure system. To appear in this category, the author acknowledges that there is no direct or institutional relationship with the trial sponsor as defined in the ACC/AHA Disclosure Policy for Writing Committees. ‡No financial benefit.

ACC = American College of Cardiology; AHA = American Heart Association.